

A Population-Average, Landmark- and Surface-based (PALS) atlas of human cerebral cortex

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This report describes a new electronic atlas of human cerebral cortex that provides a substrate for a wide variety of brain-mapping analyses. The Population-Average, Landmark- and Surface-based (PALS) atlas approach involves surface-based and volume-based representations of cortical shape, each available as population averages and as individual subject data. The specific PALS-B12 atlas introduced here is derived from structural MRI volumes of 12 normal young adults. Accurate cortical surface reconstructions were generated for each hemisphere, and the surfaces were inflated, flattened, and mapped to standard spherical configurations using SureFit and Caret software. A target atlas sphere was generated by averaging selected landmark contours from each of the 24 contributing hemispheres. Each individual hemisphere was deformed to this target using landmark-constrained surface registration. The utility of the resultant PALS-B12 atlas was demonstrated using a variety of analyses. (i) Probabilistic maps of sulcal identity were generated using both surface-based registration (SBR) and conventional volume-based registration (VBR). The SBR approach achieved markedly better consistency of sulcal alignment than did VBR. (ii) A method is introduced for ‘multi-fiducial mapping’ of volume-averaged group data (e.g., fMRI data, probabilistic architectonic maps) onto each individual hemisphere in the atlas, followed by spatial averaging across the individual maps. This yielded a population-average surface representation that circumvents the biases inherent in choosing any single hemisphere as a target. (iii) Surface-based and volume-based morphometry applied to maps of sulcal depth and sulcal identity demonstrated prominent left–right asymmetries in and near the superior temporal sulcus and Sylvian fissure. Moreover, shape variability in the temporal lobe is significantly greater in the left than the right hemisphere. The PALS-B12 atlas has been registered to other surface-based atlases to facilitate interchange of data and comparison across atlases. All data sets in the PALS-B12 atlas are accessible via the SumsDB database for online and offline visualization and analysis.

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Introduction

Atlases play an increasingly important role in providing a spatial and structural framework for visualizing and analyzing many aspects of brain structure, function, and development (Van Essen, 2002; Toga and Thompson, 2002; Mazziotta et al., 2001). In recent decades, numerous human brain atlases have been developed, each with its particular advantages and attractive features but also with various limitations and drawbacks.

For human cerebral cortex, the major challenges in generating a suitable atlas arise from the fact that cortical convolutions are very complex and are highly variable from one individual to the next. To address these challenges, strategic choices must be made on four major fronts: (i) representing cortical morphology and its variability; (ii) visualizing experimental data and underlying cortical structure; (iii) selecting coordinate systems; and (iv) choosing algorithms for registering data from individuals to the atlas. On each of these fronts, a key issue is whether the approach is based on surfaces, volumes, or a combination of both. (i) *Representing morphology and individual variability.* The core of a brain atlas is a representation of brain morphology derived from one or more individuals, captured by in vivo imaging or postmortem histological methods. Structural magnetic resonance imaging (sMRI) is particularly attractive as a substrate for a human cortical atlas because it can be readily obtained in a population of normal subjects, and it allows for isotropic volume representations that reveal the full complexity of cortical folds. Some human cortical atlases are based on an individual brain or hemisphere, which allows accurate representation of cortical shape in the selected individual but fails to represent variability across the population (Talairach and Tournoux, 1988; Roland et al., 1994; Van Essen and Drury, 1997; Geyer et al., 2001). Other atlases are based on averaging structural MRI volumes across a population of subjects, which capture some shape consistencies across the population, but invariably lead to extensive blurring of cortical convolutions (Evans et al., 1994; Friston et al., 1995; Mazziotta et al., 2001). (ii) *Visualization and analysis options.* Cortical structure is sufficiently complex that any single visualization mode is inadequate when used in isolation. It is particularly valuable to be

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able to view atlas data using a combination of volume and surface visualization. (iii) *Coordinate systems*. Spatial coordinates provide a concise and objective way to represent locations in human cortex. The most basic distinction is between 3D stereotaxic (x, y, z) coordinates and surface-based coordinates (e.g., latitude and longitude on a spherical map). Within each type, numerous specific coordinate systems are in common use, and these differ from one another in important respects. (iv) *Registration algorithms*. The objective of registering individual brains to an atlas is to compensate for multiple types of variability, including the initial position and orientation of the individual brain, the overall dimensions of each hemisphere, the pattern of cortical convolutions, and the relationship of functional subdivisions (cortical areas) to these convolutions (Toga, 1999). Given the nature of cortical structure and function, it is not possible to remove all variability, i.e., to place all geographic and functional subdivisions into precise correspondence throughout an atlas. Volume-based registration (VBR) algorithms are fundamentally limited because they fail to respect the topology of the cortical sheet and hence can lead to topologically incorrect mappings. Surface-based registration (SBR) avoids this limitation and allows more flexible application of relevant geographic and/or functional constraints, using explicit landmarks (Van Essen et al., 2005) or using metrics that reflect shape characteristics throughout the cortical sheet (Fischl et al., 1999b). However, there are as yet few evaluations of different SBR methods using objective performance measures (Desai et al., 2005).

The present report introduces the Population-Average, Landmark- and Surface-based (PALS) approach and illustrates attractive solutions it offers for each of the aforementioned challenges in generating human cerebral cortical atlases. The specific atlas illustrated here is based on structural MRI volumes from 12 normal subjects and is accordingly identified as the PALS-B12 atlas to distinguish it from other PALS atlases derived from a different set of brains. The PALS-B12 atlas data set includes (i) structural MRI volumes from individual brains and the population average; (ii) cortical segmentations and fiducial surface reconstructions for all contributing hemispheres; (iii) inflated, flattened, and spherical configurations for each hemisphere; (iv) all hemispheres registered to a target atlas surface whose landmarks reflect shape characteristics of the population rather than any individual; (v) individual surfaces resampled to a standard mesh to facilitate simultaneous visualization; (vi) multiple coordinate systems, both stereotaxic and surface-based, for expressing cortical location; (vii) maps of sulcal depth and other shape characteristics for individuals and group averages; (viii) maps of sulcal identity for individuals and the population average; and (ix) many additional types of experimental data mapped from other hemispheres to the PALS-B12 atlas.

Central to the PALS approach is the choice of geographic landmarks used to constrain the registration from individual hemispheres to the atlas. The PALS atlas uses only six well-defined and consistent geographic landmark contours, because other gyral and sulcal features in human cortex are so variable as to be unsuitable (or at least problematic) for use as landmarks. The effectiveness of this approach is demonstrated here using a variety of quantitative analyses related to sulcal shape and sulcal identity. One set of analyses relates to the issue of asymmetries between the left and right hemispheres. Although there is an extensive literature on left–right morphological asymmetries (Toga and Thompson, 2003; Ochiai et al., 2004; see Discussion), much remains

unresolved about the nature and magnitude of these differences. The surface-based morphometry (SBM) methods developed in the present study provide valuable additional information regarding left–right asymmetries, especially in the temporal lobe. They are complementary to other SBM analyses (MacDonald et al., 2000; Chung et al., 2003; Salat et al., 2004) that have focused on comparing maps of cortical thickness in different groups rather than the pattern of cortical convolutions.

Surface-based atlases are especially useful for analyzing and comparing complex functional activation patterns obtained in many thousands of fMRI studies involving a wide variety of behavioral tasks and analysis paradigms. Currently, the great majority of fMRI studies obtain statistical power by averaging across subjects using volume-based registration. Mapping such volume-based population-average data onto an individual atlas surface facilitates visualization and comparisons across studies (Van Essen, 2002), but it inevitably introduces significant biases related to the particular convolutions of the target hemisphere. Here, a method for circumventing this bias is introduced, in which fMRI data are mapped separately onto each individual fiducial surface in the PALS-B12 atlas population, then averaged across all maps. This yields an objective representation of complex activation patterns and avoids the bias of a particular individual atlas target.

All of the data sets used in this study are accessible from the SumsDB database (<http://sumsdb.wustl.edu:8081/sums>). These data can be readily visualized online (using WebCaret) and can be downloaded for offline visualization and analysis (using Caret software). A ‘scene display’ option allows immediate viewing (in either Caret or WebCaret) of the precise screen displays used in generating each of the Caret-derived panels of each figure in this paper.

Methods

Structural MRI volumes were obtained from 24 normal human subjects, all right-handed young adults (ages 18–24; data from Buckner et al., 2004; Head et al., *in press*). Data from six males and six females were used for the PALS-B12 target atlas. Multiple (three or four) T1-weighted MP-RAGE scans (1.5 T Siemens Vision scanner, $1 \times 1 \times 1.25$ mm voxels) were acquired from each individual. Volumes were registered to the Washington University 711-2C version of stereotaxic atlas space. The 711-2C target atlas is an average structural MRI generated from 6 young adult and 6 non-demented old adults; registration was a multi-stage process that resulted in a 12-parameter affine transform (Buckner et al., 2004; Ojemann et al., 1997). After registration, volumes were resampled to 1 mm^3 voxels. Spatial non-uniformity correction (intensity normalization) was applied using FSL’s FAST algorithm (<http://www.fmrib.ox.ac.uk/fsl/fast/>) for Cases 2–6 and the 3dUniformalize algorithm (AFNI, <http://afni.nimh.nih.gov/afni/>) for Case 1. For Cases 7–12, high-quality segmentations were generated without using non-uniformity correction.

The main processing sequence was done using SureFit 4.45 software for segmentation and initial surface reconstruction (<http://brainvis.wustl.edu/surefit>) and Caret 5.11 software for subsequent stages (<http://brainvis.wustl.edu/caret>). Fig. 1 illustrates several key stages in the segmentation and initial surface reconstruction of an example hemisphere (Case 1 right hemisphere). A coronal slice through the structural MRI volume (Fig. 1A) illustrates the typical image quality encountered in this data set. The SureFit method

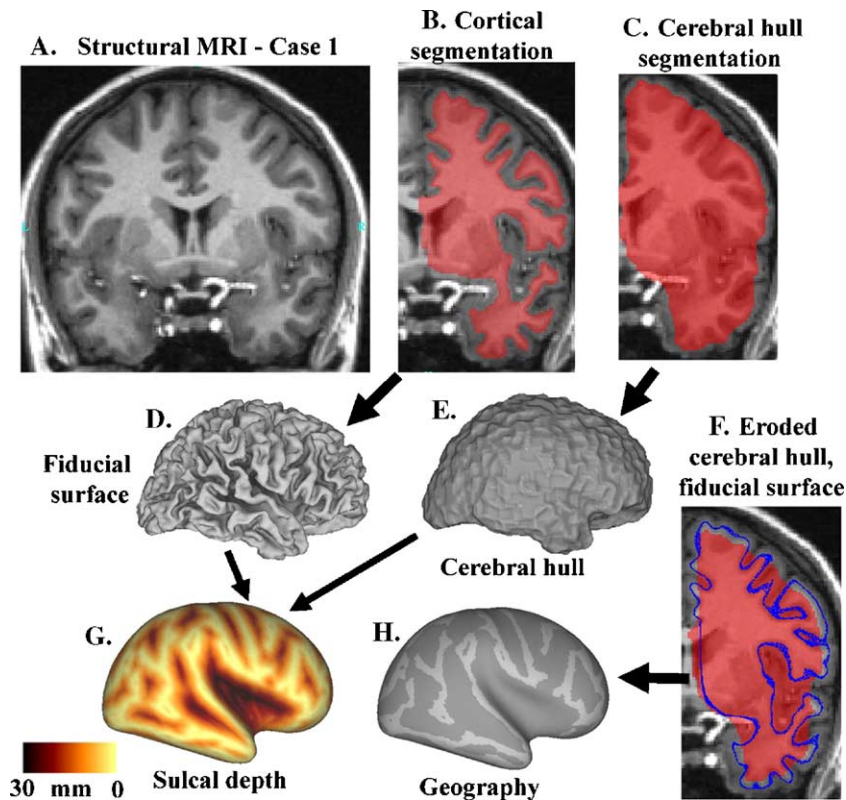


Fig. 1. Initial steps in segmentation, surface generation, and sulcal depth analysis. (A) Coronal section through the structural MRI volume for Case 1 at the level of the anterior commissure ($y = 0$). (B) SureFit-derived segmentation (pink) overlaid on the structural MRI volume for the right hemisphere. (C) Cerebral hull segmentation used for computing sulcal depth. This was generated by applying multiple dilation operations to the initial cortical segmentation (filling in all sulci but also expanding the external boundary), followed by the same number of erosion operations. (D) Lateral view of the fiducial surface for Case 1.R, generated by a tessellation around the boundary of the SureFit cortical segmentation. (E) Cerebral hull surface generated by a tessellation around the boundary of the cerebral hull segmentation. (F) An eroded cerebral hull segmentation, generated by three erosion iterations (1 mm = voxel/iteration) of the cerebral hull. A slice through the fiducial surface (blue contour) is superimposed. The eroded segmentation boundary defines the boundary between buried cortex (surface nodes within the eroded cerebral hull) and gyral cortex (nodes lying outside). (G) Sulcal depth map displayed on a lateral view of the Case 1 right hemisphere. (H) A map of cortical geography displayed on a lateral view of the very inflated surface. Data are available at <http://sumsdb.wustl.edu:8081/sums/directory.do?dirid=6259364>.

generates a segmentation boundary that runs approximately mid-way through cortical gray matter (cortical layer 4), as shown for the corresponding slice through the right hemisphere (Fig. 1B). Consequently, each square millimeter of surface area represents approximately the same cortical volume independent of gyral/sulcal location (Van Essen and Maunsell, 1980), whereas methods that generate surfaces running along the gray–white boundary tend to under-represent gyral crowns and over-represent sulcal fundi. Topological errors in the initial segmentation were eliminated by an automated error correction process followed by manual editing to eliminate residual handles and other irregularities. A ‘raw’ surface running along the boundary of the segmentation was generated and smoothed slightly to create a ‘fiducial’ surface that is the best available representation of the actual cortical shape (Fig. 1D). A segmented volume representing the cerebral hull (Fig. 1C, see figure legend for details) yielded a mostly convex segmentation boundary that runs along gyral margins without dipping into sulci. A surface generated from this cerebral hull segmentation (Fig. 1E) was used in generating maps of sulcal depth (see below).

The quality of the SureFit-generated cortical segmentations was evaluated by visual inspection of segmentation boundaries and of surface contours overlaid on the anatomical volume. Surfaces typically ran close to the mid-cortical thickness and only occa-

sionally strayed into white matter or outside the pial surface, suggesting that the surfaces are generally accurate to within about 1 mm of their desired trajectory.

Measures of cortical shape

Maps of sulcal depth, cortical folding (mean curvature), and cortical geography (buried vs. exposed cortex) were automatically generated for each hemisphere. Sulcal depth was computed as the 3D distance from each node in the fiducial surface (Fig. 1D) to the nearest point on the cerebral hull surface (Fig. 1E). Depths were computed as a direct linear distance, unconstrained by local surface geometry. Fig. 1G shows the sulcal depth map for Case 1, displayed on a lateral view of the highly inflated right hemisphere. Darker shades represent greater depths below the surface (see scale bar). Sulcal depth is a particularly useful measure of cortical shape because it provides a continuously varying measure of both coarse and fine shape attributes (Van Essen, 2004a). The sulcal depth measure used here is qualitatively similar but quantitatively different from the average convexity measure of Fischl et al. (1999a) which reflects the integrated distance each node moves during their inflation/smoothing process. It is quantitatively similar to that used by Lohman et al. (1999) to encode depth along the

fundus of each sulcus, but here all points on the surface are assigned depth values.

Maps of cortical geography were computed by intersecting the fiducial surface (blue contour in Fig. 1F) with an eroded version of the corresponding cerebral hull segmentation (pink in Fig. 1F). Nodes inside the eroded hull segmentation were classified as sulcal (buried); nodes outside this segmentation as gyral. Fig. 1H shows the geography map on a lateral view of the Case 1 right

hemisphere. An additional measure of cortical shape is folding (mean curvature; Drury et al., 1996) which emphasizes purely local shape characteristics (sulcal and gyral folds) independent of depth within the hemisphere (cf. Figs. 2A–C).

Each fiducial surface was translated to 711-2C space by placing the origin at the anterior commissure (rather than at the corner of the cropped volume used for segmentation of the hemisphere). An inflation algorithm was used to generate moderately inflated (300

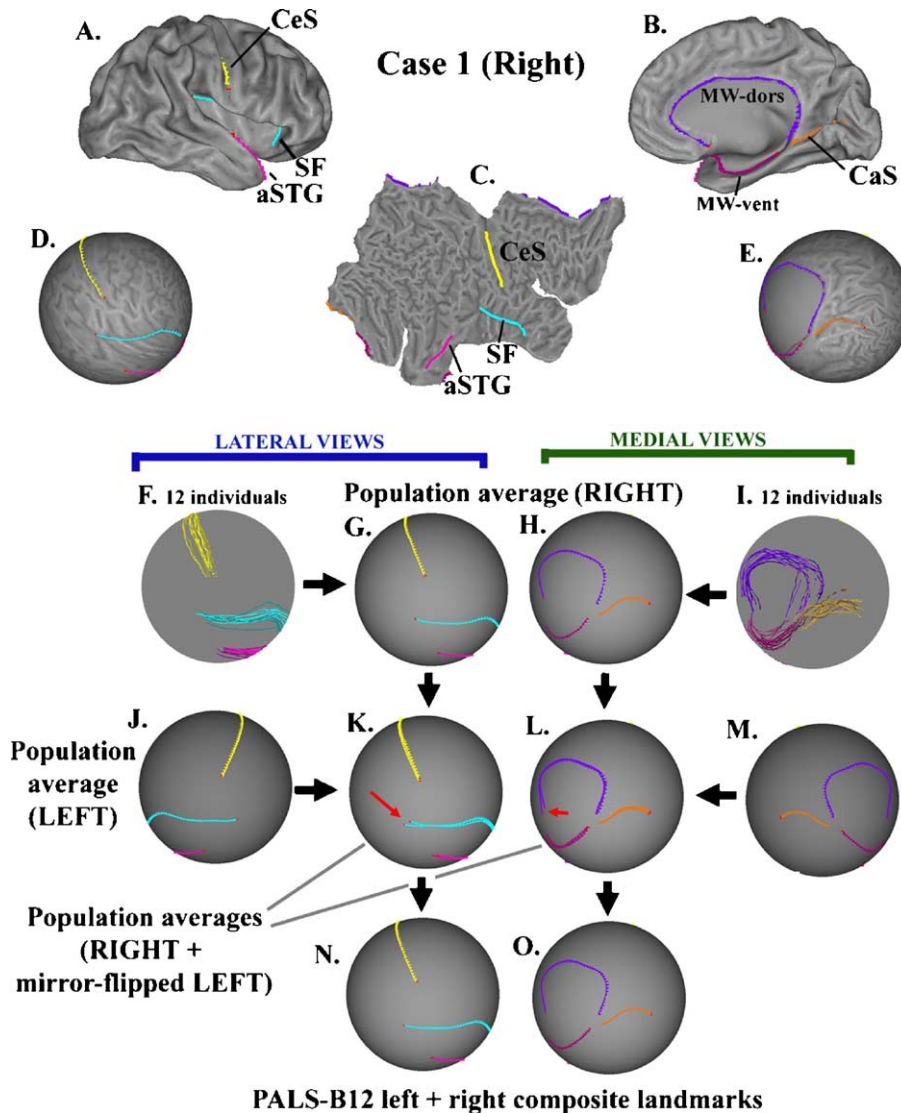


Fig. 2. Core 6 landmarks (individual hemisphere and population average). (A) Landmarks visible on a lateral view of the moderately inflated Case 1 right hemisphere: the central sulcus (CeS), Sylvian Fissure (SF), and anterior half of the superior temporal gyrus (aSTG, labeled SF_STSant in Caret for historical reasons). (B) Landmarks visible on a medial view of the moderately inflated right hemisphere: the calcarine sulcus (CaS) and the dorsal and ventral parts of the medial wall boundary (MW-dors, MW-vent). (C) Sulcal landmark contours were initially drawn on the flat map of the individual hemisphere, using cortical folding to visualize sulcal and gyral creases and were then projected onto the other surface configurations. (D) Lateral view of the spherical standard surface. (E) Medial view of the spherical standard surface. (F–O) Generation of population-average landmarks. (F) Landmark contours for the right hemispheres of Cases 1–12, saved in the spherical standard configuration for each individual and then combined and displayed on a lateral view of a target sphere. (G, H) Lateral and medial views of the population-average contours for the 12 right hemispheres. (I) Medial view of the landmark contours for the 12 individual right hemispheres. (J, M) Lateral and medial views of landmark contours for the 12 individual left hemispheres. (K, L) Lateral and medial views of the population-average contours for the 12 right hemispheres and for the 12 mirror-flipped left hemispheres. The two sets of contours overlap extensively and are readily discriminable in only a few places (e.g., red arrows). (N, O) Lateral and medial views of the combined population-average contours combined for the right and mirror-flipped left hemispheres. The average landmark contours for the female and male subgroups were computed separately; the differences averaged 4.4° (6.7 map-mm) for the left hemisphere and 8.4° (12.9 map-mm) for the right hemisphere. For details regarding the precise extent of each landmark and how to draw them on individual hemispheres, see http://brainvis.wustl.edu/help/landmarks_core6/landmarks_core6.html. Data are available at <http://sumsdb.wustl.edu:8081/sums/directory.do?dirid=6260163>.

inflation iterations) and highly inflated (1000 iterations) configurations. Additional intermediate configurations included an initial spherical configuration and a ‘compressed medial wall’ configuration that allowed key landmarks used in flattening to be seen in a single surface view. The natural boundaries of the cortex along the medial wall (adjoining the corpus callosum, basal forebrain, and hippocampus) were delineated by manual tracing on the compressed medial wall map. All nodes enclosed by this contour were designated as non-cortical medial wall nodes.

Multi-resolution morphing algorithms (Drury et al., 1996) were used to reduce areal and angular distortions on the spherical map and the flat map. For the spherical map, the reference surface was the fiducial configuration after extensive smoothing of the non-cortical medial wall region (to eliminate artifactual irregularities). The surface area of the sphere was matched to that of the reference fiducial surface. For the flat map, distortions were reduced by making cuts along five standardized trajectories relative to geographic landmarks. Flat maps and spherical maps were aligned to standard spaces (‘Cartesian standard’ and spherical standard, respectively), in both cases using the ventral tip of the central sulcus as origin and the orientation of the central sulcus as a second constraint (Drury et al., 1999).

Surface-based registration

A set of six landmarks (the ‘Core 6’ landmarks) was used for surface-based registration. These included the fundi of the calcarine sulcus, central sulcus, and Sylvian fissure; the anterior half of the superior temporal gyrus (STG); and the medial wall cortical margin (split into dorsal and ventral portions). These landmarks were selected on the basis of their consistency in location and extent (see Results); they are similar to the landmarks used in Van Essen (2004a) but differ slightly because each landmark was restricted to regions of highest consistency for each sulcus rather than extending the entire sulcal length (see Figs. 2A–C and http://brainvis.wustl.edu/help/landmarks_core6/landmarks_core6.html for details). To delineate landmarks, inflated surfaces of the individual hemisphere (Figs. 2A, B) and the corresponding Colin atlas (not shown) hemisphere were viewed side by side. Landmark positions were delineated on the inflated map (by highlighting selected nodes), drawn as explicit contours on the flat map (Fig. 2C), and projected onto the surface in a ‘barycentric’ format so that they could be viewed on any configuration, including the spherical standard surface used for registration (Figs. 2D, E).

Population-average landmarks

Population-average landmarks were generated in a sequence of steps illustrated in Fig. 2. The initial step was to project the landmark contours from the 12 contributing individuals to a ‘standard mesh’ atlas sphere (73,730 regularly tessellated nodes), as illustrated for the right hemisphere in Fig. 2A (lateral view) and Fig. 2D (medial view). Average landmark contours were then generated for the right (Figs. 2B, C) and left hemispheres (Figs. 2E, H). After mirror-flipping the left hemisphere contours, the differences between average left and right landmark contours differed only slightly (Figs. 2K, L). More specifically, the separation between corresponding points on the right vs. mirror-flipped left landmark contours differed by an average of 2.5° (3.9 atlas map-mm) on the sphere (range 1.5°–4.9° for the averages of the six individual contours). [Note: atlas map-mm refers to the distance on

a sphere whose total surface area equals the average of the contributing fiducial configurations (cf. Drury et al., 1996). Owing to distortions, 1 map-mm on the spherical map does not generally correspond to 1 mm on any of the contributing fiducial surfaces.] Hybrid left–right PALS-B12 landmark contours were generated by averaging right and mirror-flipped left average landmark contours (Figs. 2N, O). To assess the dependency on the particular choice of brains, average landmark contours were generated for a separate population of 12 normal right-handed individuals in the same age range (Cases 13–24). The resultant average landmark contours differed only slightly from the original set (mean 2.1°, range 1.2°–3.6°).

Each hemisphere was registered to the PALS-B12 left–right composite target atlas landmarks, using a spherical registration algorithm in Caret with a standardized set of parameters. This algorithm uses multiple cycles of landmark-constrained smoothing, coupled with shape-preserving morphing iterations that reduce local linear and angular distortions (Van Essen et al., 2001, 2004a,b). In situations where comparisons between groups are to be restricted to the same hemisphere (e.g., shape comparisons between the left hemispheres of group A vs. group B and a separate comparison between the right hemispheres of group A vs. group B), it is appropriate to register each hemisphere to a hemisphere-specific atlas target. These hemisphere-specific population-average landmarks are available but were not used for the analyses reported in this study (see Discussion).

Data files routinely mapped from the individual to the atlas during the registration process included the *surface shape* file (including sulcal depth and cortical folding) and the *paint* file (including maps of cortical geography and identified sulci). Also generated was a *deformation map* file containing all the information needed for post hoc mapping of additional data sets from the individual to the atlas. A complementary (inverse) deformation map file was also generated to allow data from the atlas to be registered to each individual.

Standard-mesh surface representations

Using a strategy introduced by Saad et al. (2004), the various surface configurations (fiducial, inflated, spherical, and flat map) for each individual hemisphere were resampled so that they were represented using the 73,730-node ‘standard-mesh’ lattice defined for the PALS-B12 spherical atlas. This entailed mapping each node in the atlas tessellation to the appropriately interpolated 3D position within a surface tile in the individual configuration. For flat maps, it was also necessary to generate a topology file for the atlas tessellation that included cuts matching those in the individual flat maps. After the resampling, one iteration of smoothing was applied to eliminate local irregularities. From inspection of the original and resampled surfaces as they intersect the structural MRI volume, the difference between the original and resampled fiducial was generally less than 1 mm. In aggregate, the shrinkage and smoothing led to an average reduction of 8% in the surface area of the resampled fiducial configuration compared to the original surface.

Sulcal identification

In Cases 1–12, all major sulci were identified in both left and right hemispheres. The assignments were made by comparing each individual hemisphere to labeled photographs of postmortem

brains in the Ono et al. (1990) atlas. While viewing the inflated and/or fiducial surface, selected nodes at key locations were highlighted. Aided by these marks, a contour enclosing the sulcus of interest was drawn on a flat map display of cortical geography. An option in Caret assigned nodes to a specified sulcus if they were enclosed by this contour and were identified as buried cortex on the geography map. Sulcal identification of necessity entailed many subjective judgments, because precise boundaries between one sulcus and another are often ambiguous, when a sulcus is fragmented into multiple segments (islands) by gyral interruptions or when neighboring major sulci are fused. Internally consistent criteria were applied as much as possible to minimize inadvertent biases that might exaggerate inter-individual or interhemispheric differences.

Each map of sulcal identity charted on individual hemisphere surfaces was registered to the PALS-B12 atlas surface. These were combined into ‘atlas paint files’ that allowed visualization as a probabilistic map and also served in subsequent quantitative analyses. In addition, the surface representations of sulcal identity were converted to volume representations (*volume paint files*) having a specified thickness (3 mm standard thickness, but 2 mm and 4 mm thicknesses used in certain analyses). Volume maps were combined across individuals to generate probabilistic volume representations of sulcal identity.

Quantitative analyses were made to evaluate the consistency with which sulci of a given identity were aligned on the atlas surface and in the atlas volume for the 12 contributing brains. For each sulcus, determinations were made for the number of surface nodes, $n(i)$, assigned to that sulcus in only a single case ($i = 1$), in exactly two cases ($i = 2$), etc., up to all 12 cases ($i = 12$). An index of surface alignment consistency, SAC, was then computed for each sulcus:

$$\text{SAC} = \sum_{i=1}^{12} (i-1)n(i)/11(N_{\text{total}}), \quad (1)$$

where $n(i)$ is the number of nodes having exactly i hemispheres assigned to that sulcus, and N_{total} is the total ‘envelope’ of nodes associated with that sulcus. This index ranges from zero (no consistency, when there is no overlap across individuals) to unity (perfect consistency, when all 12 hemispheres are in precise registration). In general, nodes near the core of the sulcus overlap extensively and hence are associated with high i values; those near the fringe overlap less and are associated with low i values. Similarly, in the volume-domain, a determination was made of the number of voxels assigned to the central sulcus in only one case, in exactly two cases, etc., up to all 12 cases. For each sulcus, an index of volume alignment consistency, VAC was computed as

$$\text{VAC} = \sum_{i=1}^{12} (i-1)v(i)/11(V_{\text{total}}), \quad (2)$$

where $v(i)$ is the number of voxels having exactly i hemispheres assigned to that sulcus and V_{total} is the total envelope of voxels associated with that sulcus. The VAC also ranges from zero (no overlap) to unity (perfect overlap).

Composite and population-average sulcal depth maps

A variety of analyses were carried out on the sulcal depth maps from groups of individuals registered to the atlas. Algebraic

operations included node-by-node computation of the average, standard deviation, and standard error for a population of individuals; and computing the difference between two group averages (node-by-node subtraction).

Another set of analysis options was based on pairwise cross-correlations between sulcal depth maps, made node-by-node across the entire hemisphere or for individual cortical lobes. (For example, the sulcal depth maps between the Case 1 and Case 2 right hemispheres had a correlation coefficient of 0.49; correlation coefficients between other pairs typically ranged between 0.4 and 0.6.) An $N \times N$ matrix of cross-correlation coefficients for all possible pairwise combinations between N cases was used as the input to multi-dimensional scaling (MDS) using ‘R’ statistical analysis software (<http://www.r-project.org/>). Two types of statistical analysis on the MDS data were carried out in order to test for differences between groups. Differences between left and right hemispheres were evaluated by calculating the distance between the two group means and comparing this to the distribution of distances after 10,000 randomization rounds (i.e., with data points randomly assigned to each group). Differences in scatter (individual variability) were evaluated by calculating the distance of each MDS data point from its group mean and comparing the distributions for the two groups using a t test (two-tailed).

Two tests were used to evaluate the statistical significance of regional hemispheric differences. One involved determinations applied to 12 geographically identified subdivisions, in which the mean depth within each subregion was calculated for each individual. Left–right differences in the means for each subregion were tested for significance using a randomization analysis and Bonferroni correction for multiple comparisons. The second involved a test–retest strategy in which the entire population was subdivided into two independent groups (e.g., Buckner et al., 1995). In the first group, regions of interest (ROIs) were identified by thresholding the sulcal depth difference map, identifying the largest subregions above threshold, and determining whether the mean depth within each subregion differed significantly for the left and right sides. Regions for which the ROI was above a criterion surface area were retested by identifying corresponding subregions in the second group and testing each for significance after Bonferroni correction for the number of ROIs tested.

Multi-fiducial mapping

An option for mapping group-average fMRI data onto each of the PALS-B12 individual fiducial surfaces and then computing the average across the maps was incorporated into Caret. For each of the 12 fiducial surfaces for each hemisphere, an example fMRI data set was mapped to the surface using the ‘enclosing voxel’ algorithm. This algorithm assigns each node the fMRI activation value for the voxel in which it resides ($2 \times 2 \times 2$ mm voxels in the example case). (Note—this is one of seven distinct mapping algorithms available in the Caret fMRI mapper.) The 12 values at each surface node were then averaged to create a multi-fiducial average map. The fiducial surfaces resampled from the initial segmentation process were in 711-2C space and hence are directly suitable for group-average fMRI data also processed in 711-2C space or in the very similar 711-2B space (see below). However, the great majority of published fMRI studies involve group-average data mapped to various other target spaces by a variety of volume registration algorithms. Accordingly, we registered each of the PALS-B12 volumes from 711-2C space to five other stereo-

taxic spaces (using the default mode for methods having multiple options) and have made them available in the Caret 5.2 release. This includes SPM99 (nonlinear transformation; <http://www.fil.ion.ucl.ac.uk/spm/>); SPM2 (nonlinear transformation); FLIRT (affine transformation; <http://www.fmrib.ox.ac.uk/fsl/flirt>); MRITOTAL (affine transformation; http://www.imaging.robarts.ca/Software/mni_software/mni_autoreg/readme.html); and AFNI (piecewise linear transformation; <http://afni.nimh.nih.gov/afni/>). For SPM99, SPM2, and FLIRT, the target volume is the MNI152 average volume (avg_152_T1). For MRITOTAL, the target is MNI 305 space (the ‘average-305’ volume provided with their software distribution. For AFNI, the target is the classical Talairach space (Talairach and Tournoux, 1988) and the piecewise linear transformation is identical to that originally used. In each case, the individual volumes were registered using the published procedure, then the individual surfaces were deformed using the deformation matrix defined by the volume registration process. For additional details, see http://brainvis.wustl.edu/help/pals_volume_normalization/.

The 711-2C atlas [the ‘combined (young + old) atlas-representative target image’ in Buckner et al., 2004] is a volume average generated from 12 young adult and 12 non-demented old adult brains. Each group of 12 brains was separately registered to the 711-2B version of Talairach stereotaxic space (Ojemann et al., 1997) by the process described in Buckner et al. (2004). The average PALS-B12 volume MRI and the original 711-2B average volume are highly similar in overall shape, in terms of the margins of cortical gray matter (generally within 1–2 mm) and in the location of major sulci that are discernible in the average volume. (The affine transformation between the two volumes was close to the identity matrix, differing by 1.5% along the y axis and much less for the x and z axes). Consequently, it is reasonable to use the PALS-B12 fiducial surfaces in 711-2C space as a target for multi-fiducial mapping of group-average volume fMRI data that have been registered to 711-2B space.

Probabilistic architectonic maps

Volume-based probabilistic maps of architectonic areas provided by the Zilles laboratory (Amunts and Zilles, 2000; Zilles et al., 2002; http://www.fz-juelich.de/ime/ime_start/) were mapped to the PALS-B12 atlas. These data sets were originally generated from quantitative analyses of postmortem histological sections (5–10 subjects contributing to the maps for each area). Cortical gray matter assigned to each area was mapped onto the postmortem structural MRI of the corresponding individual brain by a nonlinear registration algorithm, and each structural MRI (along with the architectonic maps) was registered to the ‘colin 27’ atlas brain by a linear affine transformation (Schormann and Zilles, 1998). The individual architectonic maps were then registered to two additional stereotaxic spaces (average_305 using MRITOTAL and 711-2B using FSL-flirt affine transformations). These were recombined to form probabilistic architectonic maps (in both average_305 and 711-2B space) that were mapped to the PALS-B12 atlas surface by multi-fiducial mapping.

Link to other surface-based atlases

Landmark-based registration was carried out to link the PALS-B12 atlas to two other surface-based atlases in widespread use. One target was the ‘Colin’ atlas (Van Essen, 2002), which is based on a high-resolution structural MRI of a single individual; the registra-

tion was done using the standard Core 6 landmarks. The Colin hemispheres were not included in the subsequent population analysis of sulcal depth, but they do provide a link to many other data sets that have been mapped to the Colin atlas.

They also provide standard left and right hemisphere flat map configurations that are useful for visualization. Another target was the FreeSurfer population-average atlas, based on 40 hemispheres (left hemisphere) and 19 subjects (right hemisphere) (Fischl et al., 1999b). The input data were the maps of average folding and average convexity (related to sulcal depth but based on the integrated distance nodes move during inflation). The FreeSurfer population-average map was rotated to align it with the spherical standard orientation illustrated in Fig. 2. Additional sulcal landmarks besides the Core 6 landmarks were needed to obtain good registration between FreeSurfer and PALS-B12 atlases, because the pattern of areal distortions differ markedly for spherical surfaces generated by FreeSurfer vs. Caret algorithms. Nine additional landmarks were reliably identified on both the FreeSurfer and PALS-B12 population-average sulcal depth maps.

Any data set that has been registered (by whatever means) to the Colin atlas or the FreeSurfer atlas can be quickly mapped to the PALS-B12 atlas using the existing hemisphere-specific deformation-map files that define point-to-point mappings between atlases (as long as the data of interest are converted to a standard Caret-compatible file format using various available utilities). In addition, data from individual hemispheres segmented and reconstructed using FreeSurfer, Brain Voyager, or various other software packages can be registered directly to the PALS-B12 atlas by importing the surface into Caret.

Data access and scene visualization

Several steps have been taken in order to facilitate navigation through the PALS-B12 atlas data sets along efficient trajectories despite the inherent complexity of the data (hundreds of individual files, ~500 MB of data). Each figure legend in this paper contains a hyperlink to a specific set of archives in the SumsDB database that allow visualization of all of the data associated with that figure. Selecting one of these archives allows immediate visualization (via WebCaret) or downloading (for offline visualization and analysis using Caret) of the data. The display options include visualization of individual ‘scenes’, each of which recreates the exact screen display used to generate one of the panels in the figure.

Many individual scenes (e.g., relating to each panel of a multi-panel figure) can be stored within a single ‘scene file’. For the present article, each scene has a descriptive header (e.g., ‘Fig. 1A structural MRI of Case 1’) that makes it easy to navigate from one panel to another. Once a scene is viewed, the display can be modified using standard Caret/WebCaret visualization and query options (e.g., changing viewing perspective, data overlays or underlays; and interrogating using ‘identify node’ and ‘identify voxel’ options). A tutorial guide through the PALS atlas is available at <http://sumsdb.wustl.edu:8081/sums/directory.do?dirid=6332260>.

Results

The results are presented in eight sections that address three broad topics. One topic concerns the diversity of cortical shape characteristics; the results can be viewed in a population of

individuals (Diversity of sulcal patterns section) or using population-average representations of sulcal depth (Population-based shape representations section) and sulcal identity (Probabilistic surface and volume maps of sulcal identity section). A second topic focuses on methodological considerations, including an evaluation of spatial distortions at various stages of analysis (Distortion analyses section); quantitative comparison of surface-based versus volume-based registration (Evaluation of registration methods section); and the mapping between surface and volume coordinate systems (Spatial localization and coordinate systems section). A third topic involves new analysis methods that make use of the PALS-B12 atlas. This includes a process for multi-fiducial mapping of group-average fMRI data and probabilistic architectonic maps (Mapping of volume-averaged group data section) and a strategy for comparing and analyzing differences among groups, illustrated using left–right cortical asymmetries (Analysis of hemispheric asymmetries section).

Diversity of sulcal patterns

Surface representations provide an invaluable substrate for visualizing and analyzing individual variability in cortical convolutions. A useful starting point is to compare sulcal patterns in a group of representative hemispheres whose surfaces were partially inflated, thereby exposing some of the internal organization of sulci while preserving many aspects of cortical shape. Sulcal depth maps, in which darker shades correspond to more deeply buried regions, are displayed on lateral views of inflated left hemispheres from four individuals in Fig. 3. Inspection of these four example hemispheres and of the full data set (see Fig. 3 legend) supports previous observations of large regional differences in the degree of sulcal variability (Ono et al., 1990; Riviere et al., 2002). In general, only a few sulci are highly consistent in shape. The majority of sulci show marked variability yet have important consistencies; a few regions are extremely variable in shape.

In the lateral views (left column Fig. 3), the two sulci with the most consistent location, extent, and branching pattern are the central sulcus (green arrows) and the Sylvian fissure (blue arrows). At the opposite extreme, the variability is so great in a few regions such as the occipito-temporal junction (purple arrows) that it is generally difficult to identify corresponding sulci in different hemispheres. Most sulci visible in Fig. 3 have intermediate characteristics: they are consistently identifiable in the great majority of hemispheres, but are variable in terms of interruptions, branches, overall extent, and/or trajectory relative to more stable landmarks. For example, the superior temporal sulcus (STS, red arrows) can be reliably identified in all 24 PALS-B12 atlas hemispheres; however, its dorsal and anterior–ventral extent varies markedly, and in some cases its fundus is branched or interrupted (Figs. 3E, G). In the medial views (right column), the calcarine sulcus (green arrows) is the most consistent sulcal landmark. Several other major sulci, including the cingulate sulcus (blue arrows), the parieto-occipital sulcus, and the collateral sulcus are consistently identifiable, but are variable to a degree that mitigates against their use as landmarks for registration.

Altogether, only three sulci (the central and calcarine sulci and the Sylvian fissure) were judged consistent enough in shape and extent to warrant their use as registration landmarks. Even for these sulci, their terminations were more variable than the main core of the sulcus and were therefore not included in the landmark contours. The anterior half of the superior temporal gyrus, lying

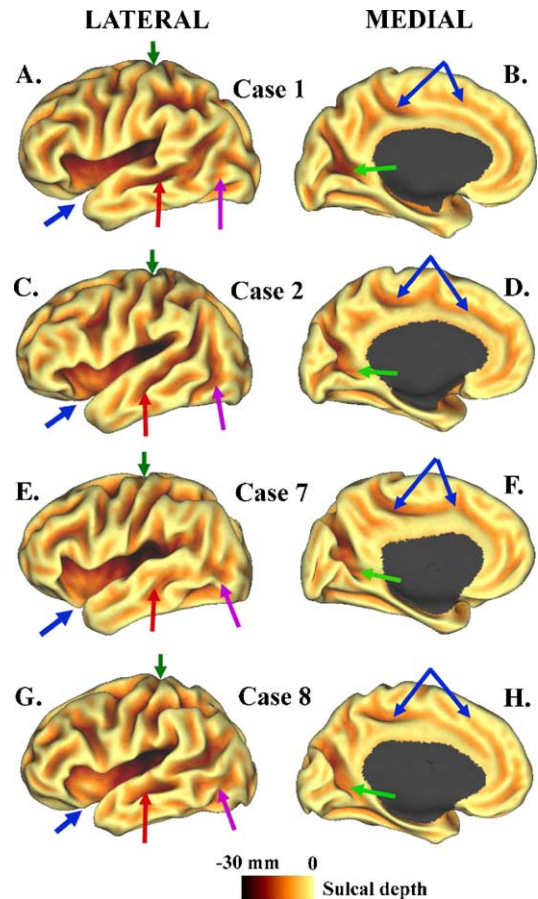


Fig. 3. Individual variability in cortical shape (lateral views). Each panel shows a lateral view (left column) or medial view (right column) of the left hemisphere from two females (Cases 1 and 2) and two males (Cases 7 and 8). Sulcal depth was computed on the fiducial surface and is displayed on moderately inflated surfaces. (A, B) Case 1. (C, D) Case 2. (E, F) Case 7. (G, H) Case 8. In left columns, Blue arrows—Sylvian fissure. Green arrows—Central sulcus. Red arrows—Superior temporal sulcus. Purple arrows—highly variable occipito-temporal junction. In right columns, Blue arrows—cingulate sulcus. Green arrows—calcarine sulcus. To view the complete set of inflated surfaces for left and right hemispheres of all 12 cases, see <http://sumsdb.wustl.edu:8081/sums/directory.do?dirid=6259376>.

between the STS and SF, is also very consistent and was used as an additional landmark (cf. Fig. 2). Finally, the margin of the cortical sheet along the medial wall was subdivided into dorsal and ventral portions and used as the remaining two Core 6 landmarks. While each of these choices is justified by an objective assessment of sulcal variability, alternative and/or additional geographic landmarks can be used to generate different atlases within the general PALS framework (see Discussion).

Distortion analyses

Changes in surface configuration (e.g., from fiducial to inflated or spherical maps) unavoidably result in complex patterns of areal distortion (compression or expansion) and linear distortions (shear). It is important to consider how and to what extent such distortions impact various analyses of data mapped to the atlas. To assess this issue, Figs. 4 and 5 provide quantitative measures of

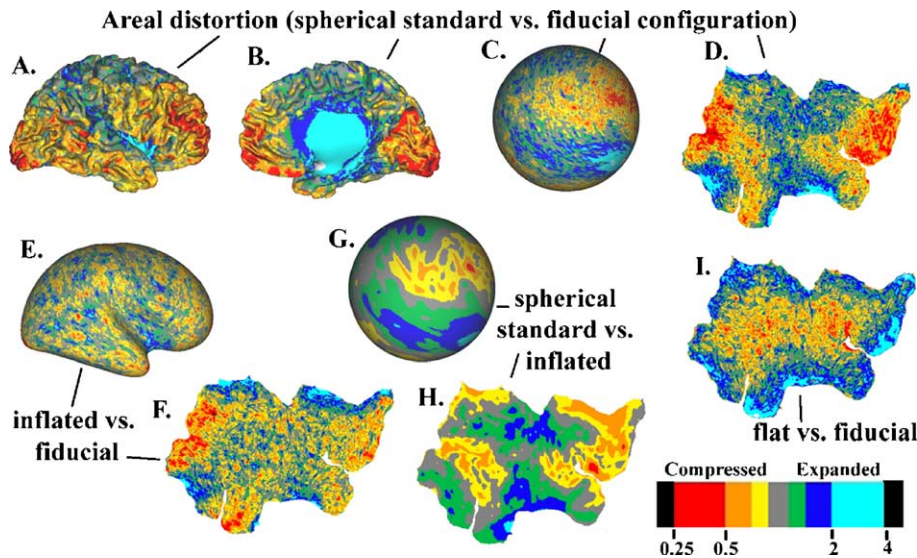


Fig. 4. Distortion maps for the Case 1 right hemisphere. On each map, compression is shown in yellow, orange, and red; expansion is shown in green and blue; and gray indicates distortion smaller than 1.15-fold (see color scale). (A–D) Areal distortion of the spherical standard map relative to the fiducial configuration, displayed on the fiducial lateral view (A), the fiducial medial view (B), a lateral view of the spherical standard surface (C), and the flat map (D). (E, F) Areal distortion of the highly inflated configuration relative to the fiducial surface, displayed on the highly inflated configuration (E) and the flat map (F). (G, H) Areal distortion of the spherical standard map relative to the very inflated surface, displayed on a lateral view of the spherical standard map (G) and the flat map (H). (I) Areal distortion of the flat map relative to the fiducial surface, displayed on the flat map. The aggregate distortion (mean of $|\log(D_i)|$, where D_i is the distortion at each node) is 0.48 for the flat map, compared to 0.49 for the spherical map. Data are available at <http://sumsdb.wustl.edu:8081/sums/directory.do?dirid=6259379>.

spatial distortion computed for key processing steps, including surface inflation, spherical morphing, flattening, and registration to the atlas sphere. Fig. 4 illustrates patterns of areal distortion associated with the inflated, spherical standard, and flat maps of the Case 1 right hemisphere. The distortion between the spherical standard and the fiducial configuration is particularly important to evaluate because the spherical standard map is a key intermediary in registration to the atlas. Accordingly, the top row of Fig. 4 displays the spherical-standard-vs.-fiducial areal distortion map on three configurations: the fiducial configuration (Figs. 4A, B, representing the denominator of the distortion ratio); the spherical standard configuration (Fig. 4C, representing the numerator); and the flat map (Fig. 4D), which is unrelated to the distortion calculation but allows the entire pattern to be seen in a single view. Regions of compression lie mainly in frontal, occipital, and lateral temporal cortex and are shown in yellow (1.15- to 1.4-fold), orange (1.4- to 2.0-fold), red (2- to 4-fold), and black (>4-fold). Regions of expansion lie mainly in the Sylvian Fissure and parahippocampal cortex (and also the non-cortical medial wall) and are shown in green (1.15- to 1.4-fold), dark blue (1.4- to 2.0-fold), light blue (2- to 4-fold), and white (>4-fold). Most of the map shows moderate distortion (63% of fiducial surface area is compressed or expanded between 1.15- and 2-fold). Much smaller portions are minimally distorted (26% in gray, corresponding to less than 1.15-fold) or highly distorted (10% of fiducial surface area compressed or expanded between 2- and 4-fold and only 3% distorted more than 4-fold).

Most of the fine-grained distortions and some of the coarser distortions arise during the initial smoothing and inflation process. This can be seen in the distortion map for the highly inflated surface compared to the fiducial surface (Figs. 4E, F). The numerous patches of local compression or expansion reflect local variations in intrinsic curvature of the fiducial surface (e.g., dimples and bumps) that are at a finer scale than the major gyral and sulcal creases. The

additional distortions that occur in mapping the highly inflated surface to the spherical standard (Figs. 4G, H) are smaller and vary more smoothly. These mainly involve compression of occipital and frontal pole regions (attributable to an elongated inflated surface being forced to conform to a spherical shape) combined with expansion of the Sylvian fissure (akin to the webbing between the thumb and fingers of a mitten being projected out to a spherical surface). Finally, it is noteworthy that the distortions of the flat map compared to the fiducial surface (Fig. 4I) are very different than those for the spherical-standard-vs.-fiducial mapping (Fig. 4D). This reflects the different geometric constraints imposed in the mapping: cuts on the flat map allow reduced compression in the frontal and occipital pole regions, but the interior of the map is generally compressed and its perimeter is generally expanded.

Registration from the individual spherical standard map to the PALS-B12 atlas sphere resulted in additional distortions whose overall magnitude was small (Fig. 5). Comparison of the sulcal depth map in the Case 1 right hemisphere before (Fig. 5A) and after registration to the atlas sphere (Fig. 5B) reveals only modest differences in the sulcal pattern. The most obvious difference is a modest rightward shift in the upper right quadrant of the spherical map (black arrows in Figs. 5A, B). These differences were quantified by (i) generating a deformation field that displays vectors between pre- and post-registration node positions for every 50th node (Fig. 5C), and (ii) computing the areal distortion between the pre-registration and post-registration spheres (Fig. 5D). The deformation field varies smoothly across the sphere, indicating an absence of major local irregularities. The areal distortion map reveals only modest distortions (mostly green and dark blue, corresponding to less than 1.4-fold expansion).

The bottom panels in Fig. 5 assess the consistency of areal distortion patterns across cases. Fig. 5E shows the aggregate atlas-sphere-vs.-fiducial areal distortion for the Case 1 right hemisphere, displayed on the Colin right hemisphere flat map. The pattern is

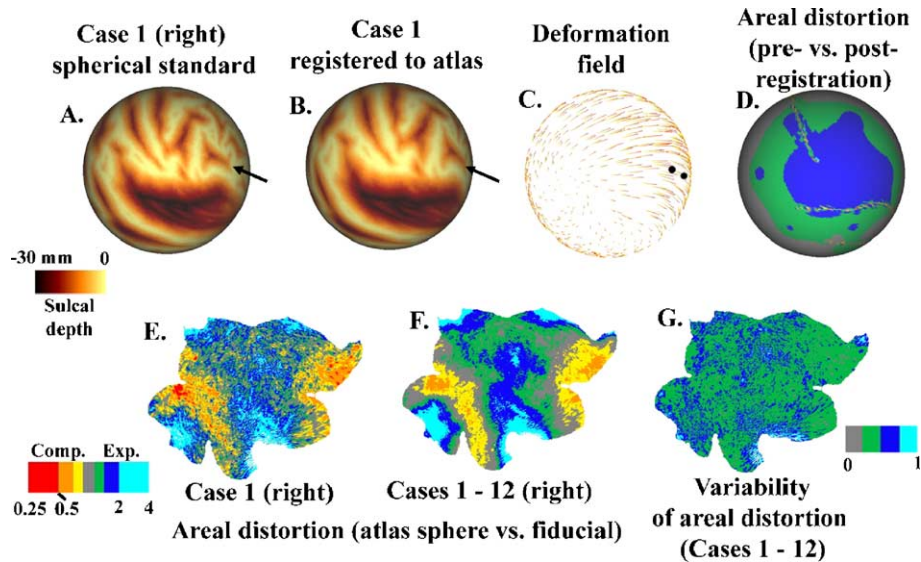


Fig. 5. Distortion associated with registration to the PALS-B12 atlas. (A) Sulcal depth displayed on a lateral view of the Case 1 right hemisphere spherical standard map. (B) Sulcal depth of the Case 1 right hemisphere after registration to the PALS-B12 atlas. (C) The deformation field between the Case 1 individual and atlas sphere, generated by displaying vectors between the pre- and post-registration node position for every 50th node on the lateral aspect of the atlas sphere. Black dots on the right show the pre- and post-registration positions of the node indicated by arrows in (A, B). (D) A map of areal distortion between the individual and atlas sphere, using the same color scale as in the preceding figure. (E) A map of aggregate areal distortion between the deformed-to-atlas Case 1 right hemisphere and its fiducial configuration, displayed on the Colin right hemisphere flat map (after registration to the PALS-B12 atlas). (F) A map of average areal distortion for the 12 atlas spheres vs. their individual fiducial surfaces for the 12 right hemispheres contributing to the PALS-B12 atlas, displayed on the Colin right hemisphere flat map. (G) A map of the standard deviation of $\log_2$ (areal distortion) for the 12 atlas spheres vs. individual fiducial surfaces for the 12 right hemispheres contributing to the PALS-B12 atlas. Data are available at <http://sumsdb.wustl.edu:8081/sums/directory.do?dirid=6259382>.

similar to that previously illustrated in Fig. 4D, but it includes the incremental distortions arising from registering the individual to the atlas sphere. The distortion map averaged across all 12 PALS-B12 right hemispheres (Fig. 5F) reveals a global pattern of compression and expansion that is strikingly similar to that for the example hemisphere (aside from many fewer fine-grained fluctuations as a result of averaging). For example, regions of compression lie mainly near the frontal and occipital poles, and the expansion in the vicinity of parahippocampal, Sylvian, central, and dorsal cingulate cortex is very similar on the individual and the population-average distortion maps. Another way to objectively evaluate this consistency is to display the variability of the group-average areal distortion (Fig. 5G). This is low in magnitude [average 0.37, computed as $\log_2$ (areal distortion)] and is relatively uniform across the map. This indicates that the regions of high compression (occipital and frontal poles) and expansion generally mapped to consistent locations on the target atlas. Thanks to this consistency, distortions in the mapping process should not be a dominant source of variability when comparing shape characteristics across hemispheres.

Population-based shape representations

The PALS-B12 atlas provides two complementary ways to visualize shape characteristics derived from the atlas population rather than just a single individual. One is to view geometric shapes whose spatial coordinates have been averaged across the population; the other is to view maps in which an explicit shape metric (sulcal depth) has been averaged across individuals. Both options are illustrated in Fig. 6 for the right hemisphere; comparisons between right and left hemisphere average maps are discussed in a later section. To generate an average fiducial surface, each of the

12 individual fiducial surfaces was resampled so that their shapes were represented using the 73,730-node standard mesh (see Methods). For each standard-mesh node, its coordinates for the 12 fiducial surfaces were averaged, yielding the average fiducial surface shown in Figs. 6A, D. This average fiducial surface is conceptually similar to the surface atlases generated by MacDonald et al. (2000) and Chung et al. (2003), but it differs somewhat in shape owing to differences in the registration methods as well as the shapes of the individual contributing hemispheres and their total number (12 in the present study vs. 150 in MacDonald et al., 2000). A similar strategy was used to generate an average for the moderately inflated configuration (Figs. 6B, E) and the highly inflated configuration (Figs. 6C, F). In each panel, the surface coloration represents a map of average sulcal depth, obtained by computing a node-by-node average of the sulcal depth maps for the 12 contributing individual right hemisphere fiducial surfaces. In regions of high consistency (e.g., the central sulcus and Sylvian fissure), the average fiducial surface (Figs. 6A, D) is reasonably similar in shape to the individual hemispheres (e.g., Fig. 1D); likewise, the average sulcal depth map is similar to that for individual hemispheres (Fig. 1G). In contrast, in regions of high variability (e.g., the occipito-temporal junction) the average fiducial surface is less convoluted than any individual hemisphere and is also irregular and bumpy. While this is predictable, given the nature of individual variability, it makes the average fiducial configuration less appealing as a substrate for visualization.

Nonetheless, this configuration remains useful as a substrate for expressing 3D spatial uncertainties and for mapping between surface and volume locations (see below). In the average sulcal depth map, regions of high individual variability are intermediate in shading (reflecting the intermediate average depth) and are blurry compared to any individual sulcal depth map.

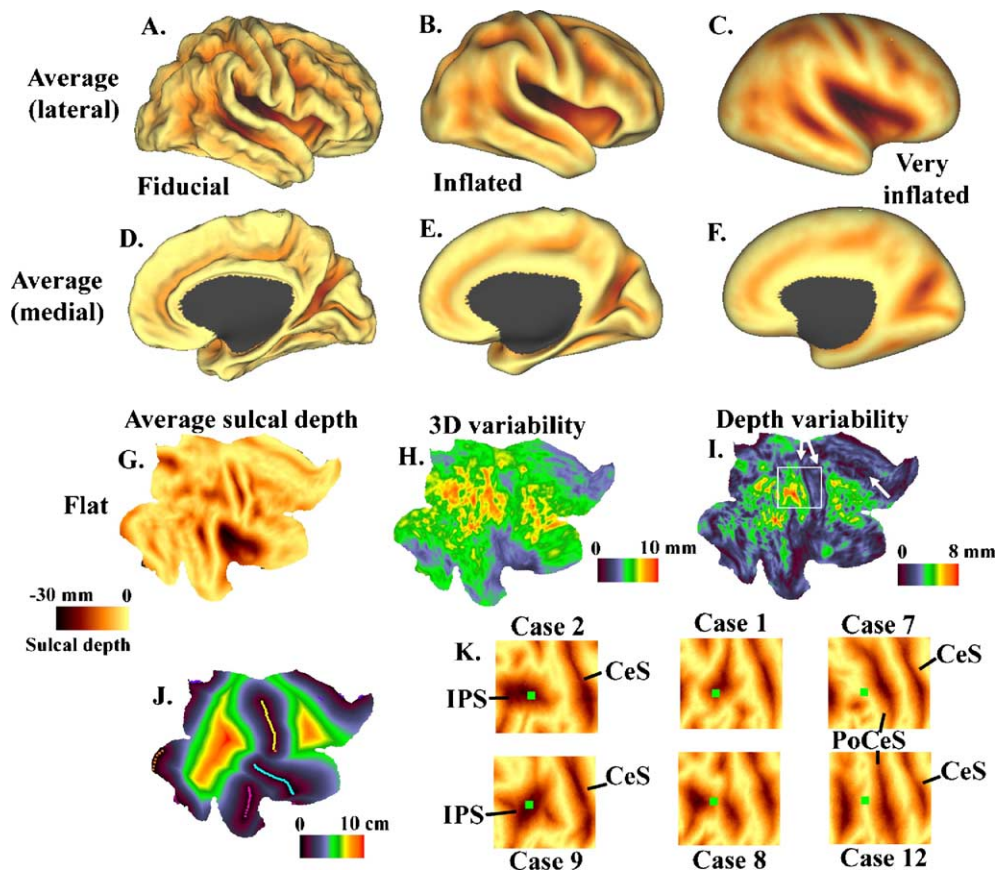


Fig. 6. Population-average representations of cortical shape. A–C. Lateral views of the average fiducial (A), inflated (B), and very inflated (C) surfaces for the Case 1–12 right hemispheres, displayed with a map of average sulcal depth from the same 12 hemispheres. D–F. Similar to A–C, but showing the medial views of the average fiducial (D), inflated (E), and very inflated (F) surfaces. G. Average sulcal depth on the Colin right hemisphere flat map registered to the PALS-B12 atlas. H. 3D variability computed across the 12 fiducial configurations. I. Depth variability (standard deviation of depth values at each node) for the 12 sulcal depth maps. J. Map of distance to closest landmark point on the sphere. K. Enlarged sulcal depth maps in the region of maximal depth variability in parietal cortex (white box in panel I) for six cases that represent very diverse patterns. Data are available at <http://sumsdb.wustl.edu:8081/sums/directory.do?dirid=6259385>.

The moderately and highly inflated average configurations are advantageous in several respects, especially when combined with displays of average sulcal depth. By their shape, the inflated configurations provide an immediate sense of approximately where one is looking on the cortical surface. The moderately inflated average map (Figs. 6B, E) retains sufficient 3D shape to see indentations corresponding to major sulci in regions of low or moderate variability (e.g., the central and superior temporal sulci on the lateral view). Informative shape characteristics are completely eroded only in regions of especially high variability (e.g., the occipito-temporal junction). The highly inflated average map (Figs. 6C, F) lacks distinct shape characteristics for all sulci except the Sylvian fissure, but it provides more uniform visibility of buried regions. In this configuration, the average sulcal depth map is even more important for being able to recognize specific sulci.

As previously noted, flat maps provide a convenient format for viewing the entire cortical surface with modest areal distortions and no perspective distortions associated with 3D surfaces viewed obliquely. Fig. 6G shows the PALS-B12 average right hemisphere sulcal depth map displayed on the Colin right hemisphere flat map configuration. The regional differences in sulcal variability evident from inspecting all 12 hemispheres were quantified using two distinct (though interrelated) measures. Fig. 6H shows a map of 3D

variability, computed as the mean distance from each node in the average fiducial surface to the corresponding nodes in each of the 12 individual fiducial surfaces. Regions of high variability (6–10 mm, in yellow and orange) are mainly in parietal cortex. Regions of low variability (2–4 mm, in gray and blue) lie mainly in the Sylvian fissure, parahippocampal cortex, and medial prefrontal cortex. Regions of moderate variability (4–6 mm, green) occupy about half of the hemisphere. Another measure of variability (Fig. 6I) is ‘depth variability’, computed as the standard deviation of sulcal depth at each location. Compared to the 3D variability in Fig. 6H, the depth variability is only about half the magnitude (average 2.9 vs. 4.3 mm; note difference in color scale), primarily because it represents differences only along a single (locally determined) axis. The difference between 3D variability and depth variability is greatest in a few streak-like regions, including the precentral and postcentral gyri and the dorso-medial convexity (white arrows). In these regions, gyri were consistently well aligned on the surface atlas and hence the variability in sulcal depth was close to zero. The greater values for 3D variability in these regions reflect the fact that the initial volume registration does not bring gyri into precise register in 3D space. Note that loci of minimum 3D variability and minimum depth variability differ from the trajectories of the landmark contours. This signifies that landmark-constrained registration preserves significant features of

individual variability even along the landmarks themselves (see Discussion).

A potential concern is that the measures of shape variability are largest in regions far from the registration landmarks, which might to some degree reflect differences in the fidelity of registration rather than intrinsic shape variability. Two additional analyses provide evidence to assuage these concerns. Fig. 6J shows a map of geodesic distance on the spherical surface to the nearest landmark point. The overall correlation between the variability patterns (Figs. 6H, I) and the distance to nearest landmark is not strong. Most of the peaks in 3D variability lie at intermediate distances from landmarks (green in Fig. 6J), and many regions that are most distant from landmarks (orange–red in Fig. 6J) are associated with intermediate depth variability (green in Fig. 6H). In addition, inspection of individual sulcal depth maps in the region of greatest depth variability reveals differences that are unlikely to reflect imperfect registration. For example, Fig. 6K shows enlarged maps of sulcal depth in posterior parietal cortex in six individual right hemispheres. In each panel, the central sulcus (CeS) is to the right and the region of maximal depth variability (9 mm SD) is near the center and is highlighted in green. In Cases 2 and 9 (left panels), the depth at this site is 24 mm, in the fundus of a prominent intraparietal sulcus (IPS) that runs mainly antero-posterior (left–right on the map). In Cases 7 and 12 (right panels), the corresponding site is at the crown of a gyrus (depth 1 mm) located between a more prominent postcentral sulcus (PoCeS) and an IPS that begins more posteriorly and (in Case 12) is oriented more medio-laterally (up–down on the map). In Cases 1 and 8 (center panels), the depth at this site is intermediate, as is the local pattern of folding. The sulcal depth maps are so diverse in this region, in terms of the location and orientation of major folds, that it would be impossible to align them all without introducing severe

local distortions. Altogether, these observations suggest that high values for 3D variability and depth variability predominantly reflect genuine biological shape variability rather than registration infidelity (see Discussion). This important conclusion is supported by visual inspection of other individual hemispheres (Figs. 4 and 5) and by additional morphometric analyses (see Analysis of hemispheric asymmetries section; Fig. 16).

Probabilistic surface and volume maps of sulcal identity

Maps of individual cortical sulci, identified using criteria from the Ono et al. (1990) atlas, were generated for 12 brains as illustrated for the right hemisphere in Fig. 7. Sulcal identities for the Case 1 right hemisphere are displayed on lateral views of the fiducial configuration (Fig. 7A), the inflated configuration (Fig. 7B), and the atlas flat map (Fig. 7C). The colors for different sulci were assigned by lobe (blue shades for temporal lobe, green for parietal, brown for frontal, reddish/orange for occipital, and purple for limbic). In the population as a whole, 18 sulci were identified in all 24 contributing hemispheres, and another 16 sulci were identified in some but not all hemispheres. The identified sulci ranged over more than a factor of 100 in surface area. (No attempt was made to explicitly identify individual gyri in this study.) The eighteen consistently identifiable sulci had an average surface area ranging from 12 cm² to 88 cm² in each hemisphere (calculated on the individual fiducial surfaces). The average fiducial surface area was 965 cm² for the left hemisphere and 962 cm² for the right; the average percentage of buried cortex was 60% in the left hemisphere and 59% in the right.

A surface-based probabilistic map of sulcal identity was generated by superimposing the maps from the individual hemispheres and determining for each node the fractional probability

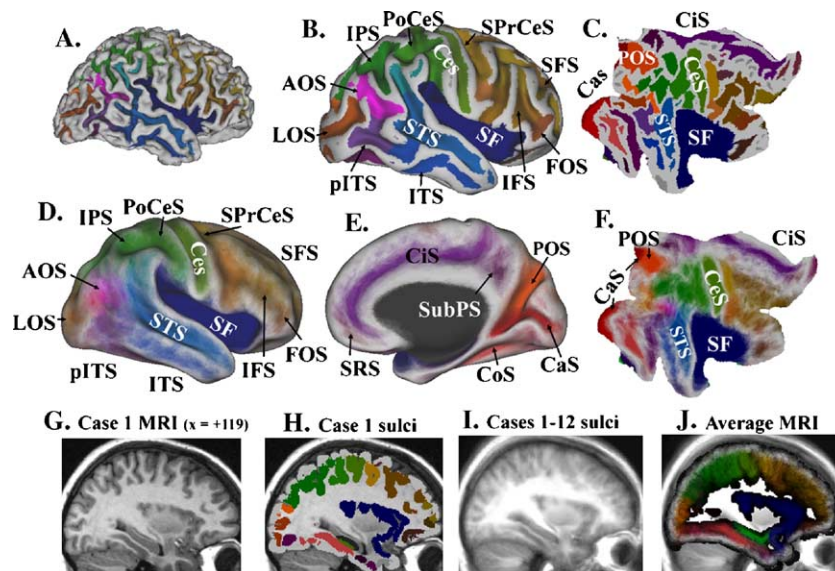


Fig. 7. Surface-based and volume-based sulcal identity maps for individuals and the population average. (A) Map of identified sulci in the Case 1 right hemisphere (lateral views). (B, C) Case 1 right hemisphere sulcal identities displayed on the PALS-B12 average moderately inflated configuration (B, lateral view) and flat map (C). See Table 1, column 1 for abbreviations. (D–F) Probabilistic map of sulcal identity for Cases 1–12 right hemispheres, displayed on lateral (D) and medial (E) views of the PALS-B12 moderately inflated configuration and on the flat map (F). (G) Parasagittal slice through the structural MRI of Case 1 right hemisphere slice 119, 30 mm lateral to the midline. (H) Map of identified sulci in the same parasagittal slice through the Case 1 right hemisphere, overlaid on the structural MRI. (I) Parasagittal slice through volume-averaged structural MRI for Cases 1–12. (J) Parasagittal slice through volume-averaged probabilistic sulcal identity map for Cases 1–12. See abbreviations list for complete sulcal names. These data, plus corresponding data for the left hemisphere, are available at <http://sumsdb.wustl.edu:8081/sums/directory.do?dirid=6259388>.

that it is associated with a particular sulcus. In the probabilistic map, shown for the right hemisphere in Figs. 7D, E (inflated configuration) and Fig. 7F (flat map), more saturated shades denote regions of higher consistency (a higher fraction of nodes having a common sulcal identity); paler shades indicate regions of greater variability in sulcal identity. In Caret and WebCaret, clicking on any given surface node in either hemisphere reports the sulcal identity as abbreviations for each of the contributing cases.

An analogous process was carried out to generate probabilistic volumetric representations of sulcal identity. A volumetric sulcal identity map was generated for each individual, yielding a sheet of 3 mm thickness in the atlas volume (see Methods). This is illustrated for a parasagittal slice through the Case 1 right hemisphere in Fig. 7H, using the same coloring as for the surface map, with gyral voxels displayed in gray. Importantly, there is excellent correspondence between this segmentation and the extent of cortical gray matter visible in the structural MRI volume (Fig. 7G); very few cortical gray matter voxels visible in the structural MRI lie outside this sulcal/gyral ribbon; conversely, few voxels in the segmented ribbon lie outside voxels that clearly appear as gray matter in the structural MRI.

Because the individual brain volumes had been registered at the outset of the study to a common stereotaxic space (711-2C space, see Methods), it was straightforward to generate a probabilistic volumetric representation of sulcal identity. Specifically, the individual volumetric maps were combined, and the fractional probability that a given voxel is associated with a particular sulcus was computed for each voxel. Fig. 7I shows a parasagittal slice through the average structural MRI volume for the 12 atlas brains. Fig. 7J shows the corresponding slice through the probabilistic volume, computed and colored in the same way as for the surface-based probabilistic atlas. Compared to the probabilistic surface map, there appears to be greater overlap between neighboring sulci in the probabilistic volume, an impression that is analyzed quantitatively in the next section.

Evaluation of registration methods

Sulcal identity maps provide an excellent substrate for a quantitative, objective comparison between Caret/PALS surface-based registration (SBR) and conventional volume-based registration. In general, consistent alignment of a given sulcus across individuals is an indication of good registration in that region; inconsistent alignment and extensive overlap among neighboring sulci are indications of poor registration. Accordingly, two different measures of registration quality were computed for both the surface and the volume maps. One is an analysis of the consistency with which each individual sulcus was aligned across the population; the other is an analysis of the degree of overlap between neighboring sulci.

Fig. 8A illustrates alignment patterns on the surface atlas for five widely separated sulci (the superior frontal, parieto-occipital, collateral, central, and superior temporal sulci). Regions of minimal consistency (only a single case representing that sulcus) are shown in black (rather than gray as in Figs. 7D–F), and regions of maximal consistency are shown with maximal saturation. The central sulcus has the highest consistency in alignment across individuals, evidenced by its predominantly bright green shading. The parieto-occipital and superior temporal sulci are intermediate, and the collateral and superior frontal show the least consistent alignment, revealed by large regions that are darkly shaded.

To analyze this issue quantitatively, an index of surface-based alignment consistency (SAC) was generated for each sulcus (see Methods). An SAC index of 1.0 would signify perfect alignment across all individuals, whereas an index of 0 would indicate no overlap whatsoever across all 12 cases. The SAC indices for the 18 largest sulci are listed in Table 1 for the left hemisphere (column 3) and right hemisphere (column 6). For sulci near landmarks, the SAC values were generally higher, as expected because they are intrinsically more consistent and are more tightly constrained by the landmarks. The SAC was largest for the Sylvian Fissure

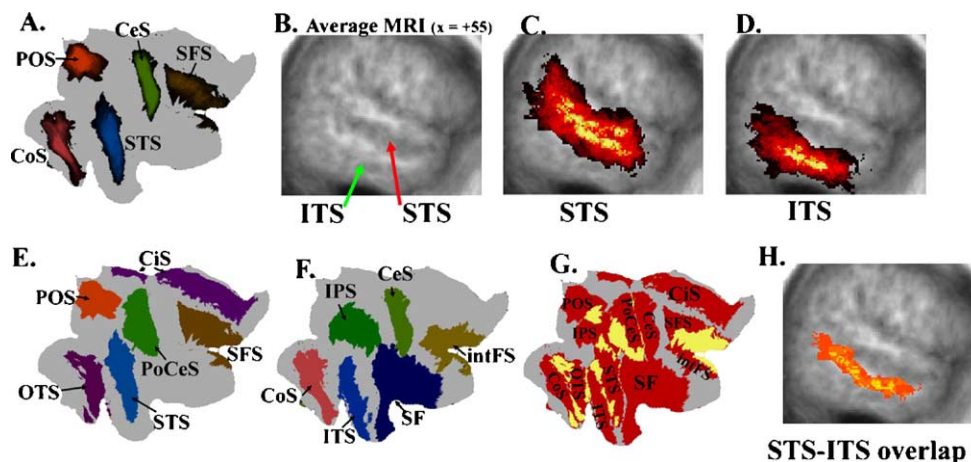


Fig. 8. (A) Map of sulcal consistency for five sulci (see Table 1 for abbreviations). For each sulcus, the darkest hue indicates locations where only one hemisphere was associated with the specified sulcus; the brightest hue indicates locations where all 12 hemispheres were consistently associated with that sulcus. (B) Parasagittal slice through the average structural MRI for Cases 1–12 (slice 144, 55 mm lateral to the midline). Arrows point to the ITS and STS. (C) Probabilistic volume map for the STS in the right hemisphere (Cases 1–12). (D) Probabilistic volume map for the ITS. (E) Thresholded probabilistic surface map for six major sulci, including three from (A) plus the PoCeS, CiS, and OTS. Thresholded probabilistic surface map for an interleaved set of six sulci, including two from (A) plus the ITS, SF, IPS, and intFS. (G) A surface sulcal overlap map for the 12 sulci shown in (E, F). Red shows locations on the atlas surface that are associated with only one of these 12 sulci in any of the 12 hemispheres mapped to the atlas. Yellow shows locations associated with multiple sulci (rarely more than two). (H) A volume sulcal overlap representation for the STS and ITS in the same parasagittal slice shown in (B–D). Data are available at <http://sumsdb.wustl.edu:8081/sums/directory.do?dirid=6259391>.

Table 1
Alignment consistency for surface and volume registration

Sulcus	Average surface area (cm ²)	Left hemisphere			Right hemisphere		
		Surface alignment consistency (SAC)	Volume alignment consistency (VAC)	SAC/VAC	Surface alignment consistency (SAC)	Volume alignment consistency (VAC)	SAC/VAC
Sylvian (SF)	88 ± 11 (SD)	0.69	0.37	1.86	0.73	0.36	2.02
Superior temporal (STS)	39 ± 10	0.45	0.25	1.83	0.49	0.27	1.79
Intraparietal (IPS)	36 ± 7	0.37	0.23	1.64	0.39	0.24	1.64
Cingulate (CiS)	33 ± 9	0.33	0.26	1.25	0.35	0.25	1.41
Central (CeS)	30 ± 4	0.53	0.29	1.83	0.56	0.25	2.21
Postcentral (PoCeS)	29 ± 5	0.41	0.24	1.70	0.40	0.23	1.73
Superior frontal (SFS)	26 ± 8	0.36	0.23	1.54	0.31	0.21	1.52
Collateral (CoS)	21 ± 4	0.39	0.19	2.05	0.40	0.21	1.94
Parieto-occipital (POS)	20 ± 6	0.44	0.25	1.74	0.43	0.27	1.60
Inferior temporal (ITS)	17 ± 14	0.20	0.16	1.28	0.22	0.18	1.26
Occipito-temporal (OTS)	16 ± 5	0.26	0.16	1.60	0.27	0.17	1.63
Intermediate frontal (intFS)	16 ± 5	0.18	0.16	1.15	0.17	0.13	1.32
Inferior frontal (IFS)	16 ± 8	0.21	0.18	1.20	0.25	0.13	1.96
Inferior precentral (IPrCeS)	16 ± 5	0.28	0.22	1.31	0.37	0.19	2.01
Superior precentral (SPrCeS)	15 ± 6	0.44	0.22	2.01	0.36	0.19	1.95
Posterior inferior temporal (pITS)	13 ± 4	0.16	0.14	1.16	0.17	0.17	0.97
Lateral occipital (LOS)	13 ± 4	0.16	0.13	1.20	0.13	0.11	1.14
Calcarine (dorsal) (CaSd)	10 ± 5	0.35	0.12	2.96	0.36	0.14	2.52
Calcarine (ventral) (CaSv)	8 ± 3	0.32	0.11	2.88	0.37	0.14	2.68
Mean		0.34	0.21	1.69	0.35	0.20	1.75

Column 1: Name of sulcus. Note that the calcarine sulcus was subdivided into dorsal and ventral subdivisions in this analysis.

Column 2: Average surface area of each sulcus ($\pm$ standard deviation), computed for the fiducial surface configuration in the left and right hemispheres for Cases 1–12.

Column 3: Surface-based alignment consistency (Eq. (1), Methods) for the left hemispheres of Cases 1–12.

Column 4: Volume-based alignment consistency (Eq. (2), Methods) for the left hemisphere.

Column 5: Ratio of SAC/VAC. Values greater than unity signify better registration quality for surface-based vs. volume-based registration.

Columns 6–8: Same as columns 3–5 except computed for right hemisphere.

Data are available at <http://sumsdb.wustl.edu:8081/sums/directory.do?dirid=6346620>.

(average 0.71) and central sulcus (average 0.56). For the group of 18 large sulci, the SAC averaged 0.34 in the left hemisphere and 0.35 in the right hemisphere. For the probabilistic volume map, the dispersion of sulci was substantially greater, as illustrated in Figs. 8C–D for the STS and ITS. In a parasagittal slice through the average structural MRI (Fig. 8B), the STS (red arrow) and, to a lesser degree, the ITS (green arrow) are readily discernible. A probabilistic map of each sulcus on its own is shown for the same parasagittal slice in Fig. 8C (STS) and Fig. 8D (ITS). A quantitative volumetric analysis was carried out by generating an index of volume-based alignment consistency (VAC) for each sulcus (analogous to the SAC index described above for the surface-based analysis). A value of 1.0 for the VAC would signify perfect alignment of sulci within the volume; an index of 0 would signify no overlap whatsoever across all 12 cases (see Methods). As indicated in Table 1 (columns 4 and 7), the VAC was largest for the Sylvian Fissure (average 0.36) and central sulcus (average 0.27); both values were about half the corresponding SAC values. For the group of 18 largest sulci, the VAC averaged 0.21 on the left and 0.20 on the right. The VAC was smaller than the SAC for all but the posterior ITS in the right hemisphere ($P < 10^{-3}$, 2-tailed t test), implying that SBR outperformed VBR throughout the cerebral cortex.

The volumetric analyses of alignment consistency are of a different dimensionality than the surface-based analyses and

depend on the thickness assigned to the cortical ribbon. Hence, it is important to consider whether the surface vs. volume differences might be artifactual, owing to an inappropriate choice of ribbon thickness. Accordingly, the analyses were repeated using a sulcal ribbon of 4 mm, which by visual inspection substantially exceeds the actual gray matter thickness. This resulted in an average 12% increase in the VAC, but the VAC remained consistently lower than the SAC (except for the pITS) to a highly significant degree ($P < 0.002$).

A second measure of registration quality is the degree to which overlap between nearby sulci occurs on the probabilistic map. Figs. 8E–G show an overlap analysis carried out on 12 major sulci that have many pairwise adjacencies. Fig. 8E shows all nodes belonging to one subgroup of sulci (POS, OTS, STS, CiS, SFS, PoCeS) in one or more cases. These sulci are non-adjacent in any individual hemisphere, and there is no overlap between them on the atlas map. Fig. 8F shows all nodes belonging to an interleaved subgroup of sulci (CoS, IPS, ITS, SF, and intFS), and again there is essentially no overlap. When the two subgroups are combined (Fig. 8G), the overlap is shown in yellow and the non-overlapping regions in red. There is virtually no overlap between some adjacent sulci (e.g., SF/STS; CeS/PoCeS). The overlap is substantial for the IPS/POCeS and SFS/intFS pairs and is intermediate for the other pairs (e.g., ITS/STS). Altogether, 17% of the 518 cm² surface area associated with these 12 sulci showed some pairwise overlap.

A similar analysis of overlap between neighboring sulci in the volume-based probabilistic map revealed considerably greater intermixing between adjacent sulci. Fig. 8H shows the overlap between the STS and ITS in the same parasagittal slice shown in panels B–D. The volume overlap between the STS and ITS was 10 cm³, or 16% of their aggregate volume (73 cm³), which is about 50% greater than the 11% overlap resulting from surface-based registration.

For the many pairwise combinations of adjacent sulci, the overlap was consistently greater for the probabilistic volume than for the probabilistic surface map. This was quantified by comparing the sum of the volumes associated with the individual sulci (474 cm³) and comparing this to the total volume associated with any of the 12 sulci analyzed (369 cm³). This ratio, 1.28, signifies an average volume overlap of 28% between adjacent sulci, which is about half again as large as the 17% overlap between adjacent sulci resulting from surface-based registration. As with the alignment consistency analysis discussed above, the volume overlap analysis depends on the thickness assigned to the cortical ribbon and might in principle be overestimated if the thickness estimate were too high. Therefore, the volume overlap analysis was repeated using a thickness of 2 mm, which by visual inspection was substantially less than the actual cortical thickness in most regions. The summed volume of the individual sulci (429 cm³) exceeded the aggregate volume associated with one or another of these sulci for the 2-mm thickness data (344 cm³), by a ratio of 1.25, which substantially exceeds the 17% overlap resulting from the surface-based analysis. Thus, the difference is not an artifact of an inappropriate thickness estimate.

In order to determine whether alternate VBR algorithms might achieve a higher registration quality, the volume analyses were repeated on probabilistic sulcal volumes registered to MNI152

space by the low-dimensional nonlinear SPM2 algorithm. The resultant VAC values were 5% higher on average for the 18 largest sulci (range –13% to +23%), but they remained significantly lower than the SAC values (average SAC/VAC_{SPM2} 1.68, $P < 0.002$). This indicates that the choice among low-dimensional VBR algorithms does not make a major difference relative to that attainable using SBR (see Discussion).

Spatial localization and coordinate systems

Besides the geographic designation of sulcal location discussed in the preceding section, additional options are available for expressing spatial locations on the PALS-B12 atlas. One involves maps of cortical areas and functionally defined subdivisions obtained from studies of other brains and registered to the atlas surface. One major source of such data derives from cortical subdivisions previously mapped to the Colin atlas (Van Essen, 2002, 2004; Van Essen et al., 2005) and transferred to the PALS-B12 atlas using the registration between these two atlas surfaces described above. For example, Fig. 9A shows the Brodmann architectonic scheme mapped from the original publication (Brodmann, 1909) onto the PALS-B12 flat map surface by way of intermediate atlas surfaces (Drury et al., 1999; Van Essen, 2002). Similarly, Fig. 9B shows a combined visuotopic scheme (Hadjikhani et al., 1998; Van Essen, 2004b) and orbito-frontal partitioning schemes (Öngür et al., 2003; Van Essen et al., 2005) that have been registered in this fashion. As additional schemes are generated and published, they can be added to this human atlas, just as has been done to populate the current macaque surface-based atlas with more than a dozen distinct partitioning schemes (Van Essen, 2004b; Van Essen et al., 2005). For volume data,

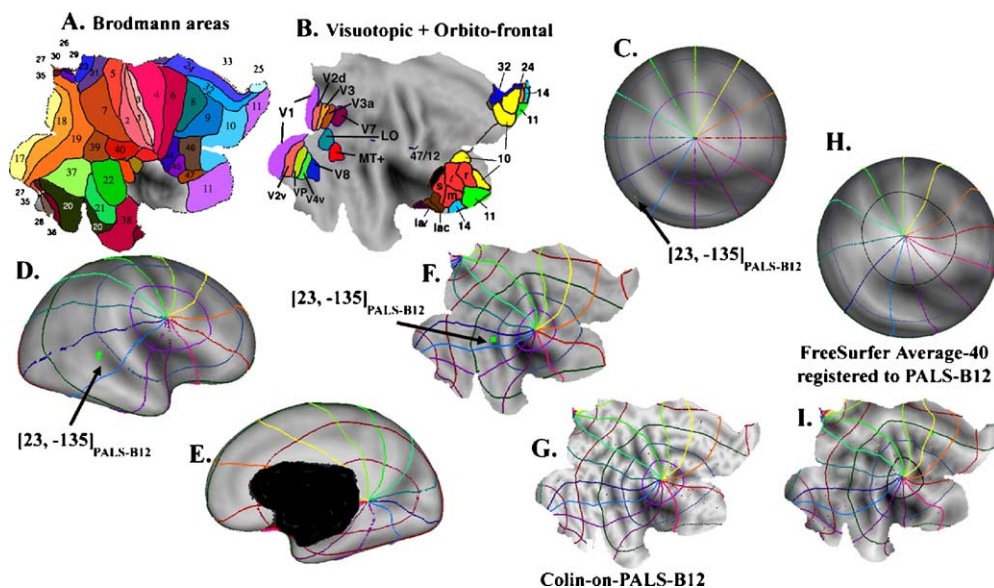


Fig. 9. Areal boundaries and coordinate systems on the PALS-B12 atlas. (A) Map of Brodmann cytoarchitectonic areas, originally mapped from Brodmann's illustration onto the Visible Man atlas (Drury et al., 1999), then registered to the Colin atlas (Van Essen, 2002) and from there to PALS-B12 and displayed on the atlas right hemisphere flat map. (B) Map of visuotopic areas (Hadjikhani et al., 1998; Van Essen, 2004b) and orbito-frontal areas (Öngür et al., 2003), registered from the Colin atlas to PALS-B12 and displayed on the atlas flat map with an underlay of average sulcal depth. (C) Latitude and longitude on the PALS-B12 atlas sphere (lateral view). (D–F) Latitude and longitude on the PALS-B12 atlas highly inflated surface (lateral and medial views) and flat map (F). Arrows point to location with coordinates [23, –135]_{PALS-B12}. (G) Latitude and longitude isocontours generated the Colin atlas sphere and displayed on the flat map. (H, I) Average curvature and latitude–longitude isocontours from the FreeSurfer 'average 19' atlas right hemisphere after registration to PALS-B12 using landmark-constrained registration and displayed on spherical (H) and flat map (I) configurations. These data, plus corresponding data for the left hemisphere are available at <http://sumsdb.wustl.edu:8081/sums/directory.do?dirid=6259397>.

including probabilistic architectonic maps, a new approach to mapping such data onto the PALS-B12 atlas is described below (cf. Fig. 12).

A complementary option is to specify locations more objectively, precisely, and concisely using spatial coordinate systems. The importance of this alternative is highlighted by the fact that most areal boundaries in human cortex are associated with major imprecision, uncertainty, error, and/or controversy. The PALS-B12 atlas supports multiple coordinate systems, both surface-based and volume-based. The primary surface-based coordinate system involves latitude and longitude values determined for the left and right hemisphere atlas spheres. A convenient way to visualize the basic layout of the spherical coordinate system is to view latitude and longitude isocontours generated on the spherical surface (shown for the right hemisphere in Fig. 9C) and projected to inflated maps (Figs. 9D, E) and flat maps (Fig. 9F). Every node on the atlas surface has an explicit latitude and longitude (encoded in a 'latlon' file) and expressed in general as $[\theta, \phi]_{\text{PALS-B12}}$.

Spherical coordinate systems have also been published for other atlas surfaces, including the Colin atlas (Van Essen, 2002) and the FreeSurfer atlas (Fischl et al., 1999a). The consistency of surface coordinates across atlases can be assessed objectively once there is a registration between them. For example, the spherical coordinates previously published for the Colin right hemisphere atlases are displayed on the atlas flat map in Fig. 9G, with the sulcal depth for the Colin hemisphere shown as an underlay. The trajectories of corresponding latitude and longitude isocontours are similar but not identical to those from the PALS-B12 atlas (Fig. 9F). These differences were quantified by calculating the average separation between corresponding latitude and longitude isocontour points for the PALS-B12 atlas and the Colin right hemisphere registered to PALS-B12. The mean difference was 12 spherical map-mm (8°) for the 5 latitude and 12 longitude isocontours (range 4.4° – 11°). These differences, while significant, are comparable in magnitude to the differences between commonly used stereotaxic coordinate systems

(see below). Similarly, Figs. 9H, I show spherical coordinates and the underlying 'average curvature' map for the FreeSurfer population-average right hemisphere atlas (19 hemispheres averaged by the spherical registration method of Fischl et al., 1999b). These are shown as a spherical map after registration to the PALS-B12 sphere and after projection to the atlas flat map. Again, the deformed latitude and longitude isocontours are similar but not identical to those directed computed on the PALS-B12 atlas (average isocontour differences = 10°).

Given the importance and widespread usage of stereotaxic coordinate systems, it is desirable to be able to relate locations on the PALS-B12 atlas surface to corresponding locations in 3D space. However, there is no deterministic, one-to-one mapping between PALS-B12 surface coordinates and stereotaxic coordinates because the PALS-B12 atlas was generated from a population of individuals; there is no reason to choose one vs. another of the 12 individual fiducial surface configurations to specify a stereotaxic location. Instead, it makes sense to express this relationship using the average fiducial configuration to specify a 'most likely' 3D location plus an explicit representation of 3D variability at that location. Caret provides an option for representing this uncertainty by displaying a 'variability sphere' whose radius equals the 3D variability metric previously illustrated in Fig. 6H. Fig. 10 shows several 3D variability spheres for different locations on the average fiducial surface (Fig. 10B) and the corresponding locations on the average highly inflated surface (Fig. 10C), the spherical map (Fig. 10E), and the volume, which is an average of the 12 structural MRI volumes (Figs. 10A, D, F). At each selected location, about half of the individual fiducial surfaces would have their corresponding nodes inside this sphere. The selected node at spherical coordinates $[22^\circ, -142^\circ]_{\text{PALS-B12}}$, in a region of high variability in the STS, has a variability sphere of radius 8 mm. The node selected in the Sylvian fissure $[26^\circ, -35^\circ]_{\text{PALS-B12}}$ has a much smaller variability sphere (3 mm); the node along the precentral gyrus $[62^\circ, 90^\circ]_{\text{PALS-B12}}$ was intermediate (6 mm variability). Of course, the distribution of

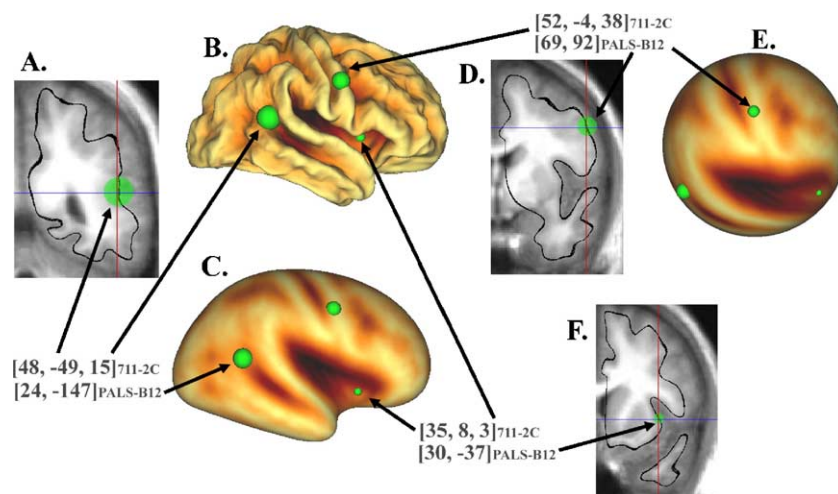


Fig. 10. Variability spheres for different locations and in different display formats for the PALS-B12 atlas. (A) Coronal slice through the population-average structural MRI, with the average fiducial surface contour (black) overlaid and with a large spatial variability sphere (green, radius 8 mm) representing the mean 3D variability associated with this location in the STS (mean distance of individual fiducial surface nodes from the average fiducial surface node). (B) Average fiducial surface (lateral view) with spatial variability spheres displayed for the STS (same location as in A), the precentral gyrus (same node as in D), and Sylvian Fissure (same node as in F). (C) Average highly inflated surface (lateral view) with the same three spatial variability spheres displayed. (D) Coronal slice intersecting a medium-sized spatial variability sphere (radius 6 mm) on the precentral gyrus. (E) Coronal slice intersecting a small spatial variability sphere (radius 3 mm) in the Sylvian Fissure. Data are available at <http://sumsdb.wustl.edu:8081/sums/directory.do?dirid=6259394>.

corresponding nodes may in many cases be anisotropic, so the spherical shape for 3D variability is only an approximation to the actual statistical distribution of any given set of corresponding nodes.

The example in Fig. 10 is based on 3D data sets in 711-2C space (the space in which the segmentations were initially carried out). However, the PALS-B12 atlas supports data registered to other stereotaxic spaces based on different target volumes (MNI152, MNI305) and/or different registration algorithms (SPM99, SPM2, FLIRT, MRITOTAL, and AFNI, see Methods). For each of these, the individual MRI volumes and the left and right hemisphere individual fiducial surfaces were registered using the published method. Given that some registration software provides multiple options for standard registration algorithms, it is important to specify this in the description of PALS-B12 atlas

surfaces and volumes (e.g., $[x,y,z]_{\text{SPM99NL}}$ for the nonlinear SPM99 transformation to MNI152 space). Depending on which atlas data set is downloaded, the average fiducial surface and associated variability map are available (PALS-B12-SPM99, etc.). The differences in 3D coordinates can be substantial—for example, in comparing 711-2C, SPM2, and AFNI registrations to stereotaxic space, the three nodes selected in Fig. 10 have stereotaxic coordinates that differ up to 6 mm, 5 mm, and 4 mm along the x , y , and z axes, respectively.

Mapping of volume-averaged group data

In the majority of published fMRI studies, data from multiple subjects are averaged to an atlas volume in order to increase statistical power. Fig. 11 illustrates one such example, in which

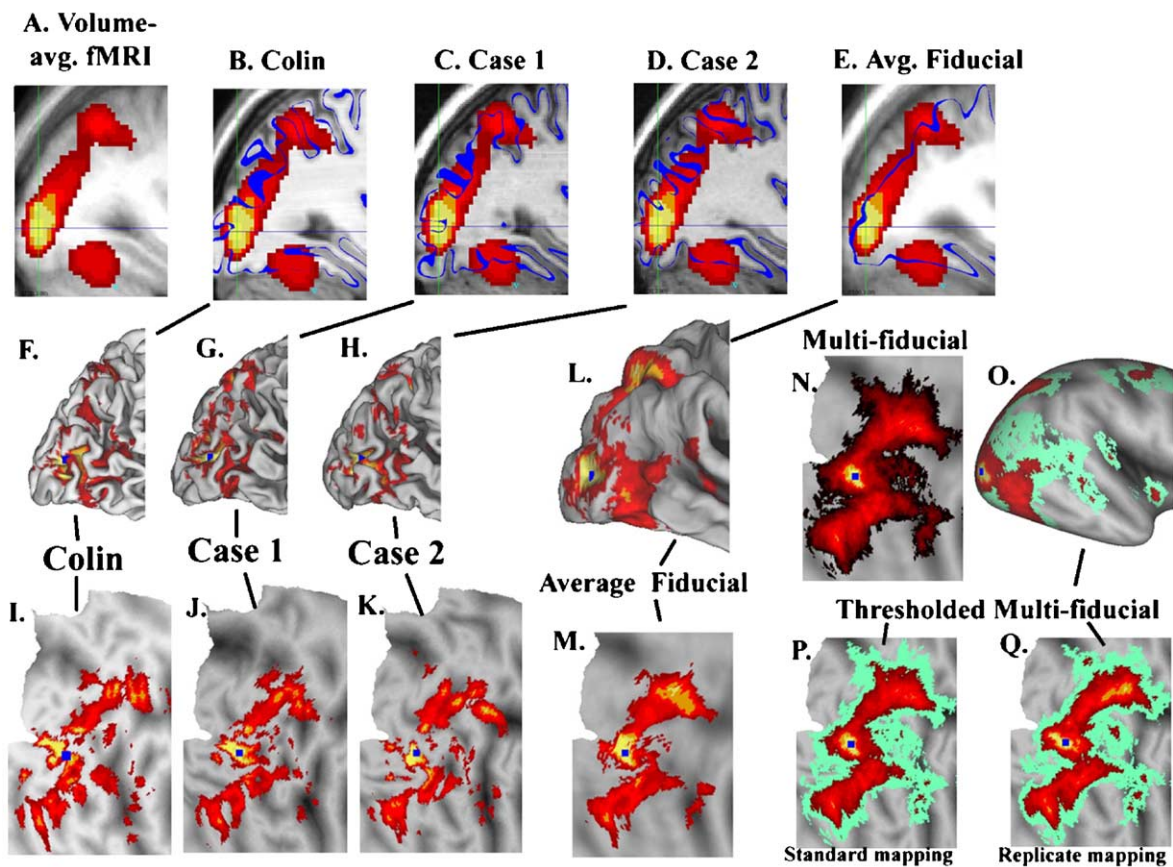


Fig. 11. Multi-fiducial mapping of volume-averaged group fMRI data. (A) Parasagittal slice through posterior cortex (28 mm lateral to the midline), showing group-averaged fMRI data ($n = 13$) from a study of attention to a spatial location (Corbetta et al., 2000), overlaid on a population-average structural MRI (711-2B) to which the data were registered. (B) The same group-average fMRI activation slice overlaid on the Colin atlas right hemisphere structural MRI and fiducial surface reconstruction contour. (C, D) The same group-average fMRI activation slice overlaid on the right hemisphere structural MRI and fiducial surface reconstruction for the Case 1 (C) and Case 2 (D) right hemispheres. (E) The same group-average fMRI activation slice overlaid on the average PALS-B12 fiducial surface contour and the average PALS-B12 structural MRI. (F–H) The group-average fMRI data mapped onto the fiducial surface reconstruction for the Colin right hemisphere (F) and the Case 1 (G) and Case 2 (H) right hemispheres. (I–K) The group-average fMRI data mapped onto the fiducial surface reconstructions for the Colin right hemisphere (I), Case 1 (J), and Case 2 (K), and displayed on the atlas flat map with the corresponding sulcal depth map underlaid. (L–M) The group-average fMRI data mapped onto the average fiducial surface reconstruction for Cases 1–12 right hemispheres (L) and displayed on the atlas flat map (M). (N) The MFM fMRI average activation, generated by spatial averaging of the mapping to the 12 individual right hemisphere fiducial surfaces. (O, P) Lateral inflated map (O) and flat map (P) views of the MFM fMRI average activation, thresholded to have the same surface area as that for mapping to the individual configurations (red, yellow), with the remaining extent of the full MFM mapping displayed in pastel green. (Q) Replicate mapping of the same fMRI volume onto a different group of 12 hemispheres. The patterns are very similar, though not identical in (P, Q). Data are available at <http://sumsdb.wustl.edu:8081/sums/directory.do?dirid=6260384>. This includes the results from different algorithms for mapping to individual hemispheres (see text in Methods and Results). Fiducial surface sets in six different stereotaxic spaces for use in multi-fiducial mapping are available at <http://sumsdb.wustl.edu:8081/sums/directory.do?dirid=6348022>.

visual attention to a spatial location (Corbetta and Shulman, 2002; Corbetta et al., 2000) activated multiple contiguous subregions in occipital, parietal, and temporal cortex. In Fig. 11A, above-threshold activation in the group-average z statistic (13 subjects) is visualized in a parasagittal slice that is overlaid on a population-averaged structural MRI atlas (711-2B). In the volume-averaged data, cortical shape information from the contributing individual subjects is not preserved, and surface reconstructions from the individuals are typically not available. In order to visualize such group-average volume data on cortical surface representations, our laboratory has previously advocated a strategy of mapping the fMRI volume data onto an individual atlas surface such as the Colin atlas (Van Essen, 2002; Van Essen et al., 2005). For example, in Fig. 11B, the same fMRI data are overlaid on the structural MRI and a corresponding surface slice (blue contour) of the Colin right hemisphere. The volume data were mapped onto this atlas surface by assigning each surface node the z -statistic value for the voxel in which it resides (the ‘enclosing voxel’ mapping algorithm). This results in a mapping to the Colin right hemisphere surface displayed on the fiducial configuration (Fig. 11F, left center) and flat map (Fig. 11I, lower left), in both cases showing only a restricted portion of occipital, temporal, and parietal cortex. Overall, the pattern is complex and patchy. Regions of near-maximal activation (yellow) map to several foci in occipital and parietal cortex. However, some aspects of this pattern reflect how the particular convolutions of the Colin surface intersect the population-averaged fMRI volume.

This distinction highlights a fundamental problem with mapping fMRI data to any individual atlas surface: the precise pattern of activation depends heavily on the shape of the target surface. This dependency is illustrated in Fig. 11 by mapping the identical fMRI volume data onto two additional surfaces—the Case 1 and Case 2 right hemisphere surfaces from the PALS-B12 atlas. Each hemisphere’s fiducial surface intersects the fMRI volume in a different way than the Colin surface (compare blue contours in Figs. 11B–D). This reflects individual variability in convolutions that are not compensated by the volume registration algorithm, and it leads to marked differences in the pattern of fMRI activation on the fiducial surface (Figs. 11F–H in the middle left) and on the flat map (Figs. 11I–K in the bottom left). The total surface areas of the three activation patterns are similar, but the patterns differ in many details. For example, the strongest occipital activation focus in the volume (yellow in the top row; note black arrow pointing to stereotaxic coordinates [33, –93, 1]_{711-2B}) maps to locations whose centers differ by more than 3 mm-cm on the flat map (black arrows in bottom row). Using the map of visual areas registered to the PALS-B12 atlas surface (Fig. 9C), this focus is centered on area LO, but the activation may extend into neighboring area MT. The other two foci lie mainly outside known visuotopic areas but may extend into area V8 ventrally and V7 dorsally.

Several options for addressing this problem are now available using the PALS framework. One is to map group-averaged data to the average fiducial surface. This is shown in a slice through the volume and surface (Fig. 11E) and on surface reconstructions (Figs. 11J, M). Average fiducial mapping (AFM) has the advantage that the target surface reflects the shape characteristics of a population, rather than just a single individual. It results in a pattern that is smoother than occurs for mapping to any individual, with only a single prominent focus in occipital cortex and another in parietal cortex. AFM also has the advantage that each surface node can be assigned the value of a single voxel (using the ‘enclosing voxel’ mapping algorithm), thereby preserving the same scale (e.g., z

statistic) as for the group-average volume data. On the other hand, a significant drawback is that the average fiducial surface has a substantially different shape than that of any individual hemisphere and is prone to its own biases. For example, the average fiducial surface contour in Fig. 11E only grazes the prominent occipital focus in this slice, whereas all of the illustrated individual fiducial surfaces intersect this focus more substantially (albeit in different places).

Multi-fiducial mapping (MFM)

An attractive alternative for addressing these issues is to map the fMRI data onto the entire population of individual fiducial surfaces in the PALS-B12 atlas family and then to generate a population-average across these surface maps (Fig. 11N). When mapped to all 12 PALS-B12 right hemispheres, the total extent of activated cortex in each individual occupied a similar surface area on the flat map ($65 \text{ cm}^2 \pm 4 \text{ cm}^2 \text{ SD}$), but as expected, the average map is spatially blurred. The averaged activation pattern occupies almost three times the surface area (183 cm^2 of the atlas map), and includes many nodes whose average value was lower than the threshold for significant z statistic ($z = 3$) in the volume data.

The multi-fiducial average map in Fig. 11N by its nature reflects information about spatial variability as well as the magnitude of the fMRI activation in the volume and is thus a hybrid representation. In Fig. 11O (very inflated surface) and Fig. 11P (flat map), this representation was thresholded so that its total extent matches the average obtained in mapping to the individual hemispheres. Regions above this threshold are most likely to have consistently contributed to the group-average volume activation and are displayed using the same color scale as in the other panels. Regions below threshold contributed to the activation inconsistently or to a lesser degree and are shown in a light green shading, to make the pattern visible but unobtrusive. To test whether MFM depends critically on the specific choice of target surfaces, the mapping was repeated for a separate group of 12 hemispheres (Cases 13–24). The results, shown in Fig. 11Q, differ only slightly from the pattern obtained with the first group of surfaces. For example, the peak MFM activations differ by only 5 map-mm in the two mappings.

A number of other mapping options are available in Caret within the MFM framework. For example, rather than displaying the average value across all 12 cases at each node, either the maximum value or the minimum value for each node can be computed and displayed. Alternative options are to use any of six other algorithms for mapping voxel values to each node in an individual surface. For example, the ‘average voxel’ method determines the average value for all voxels within a specified distance of each surface node. When this method was applied using an $8 \times 8 \times 8 \text{ mm}$ mapping volume, the mapping to each individual hemisphere was several-fold more widespread in extent, as expected. However, the MFM maps for the enclosing voxel vs. the 8-mm average voxel algorithms were very similar. This is because the blurring resulting from the larger voxel size was smaller than that associated with individual variability (data not shown). Thus, MFM is more effective at compensating for individual variability than are spatial blurring options that are constrained by mapping to a single target brain.

Probabilistic architectonic maps

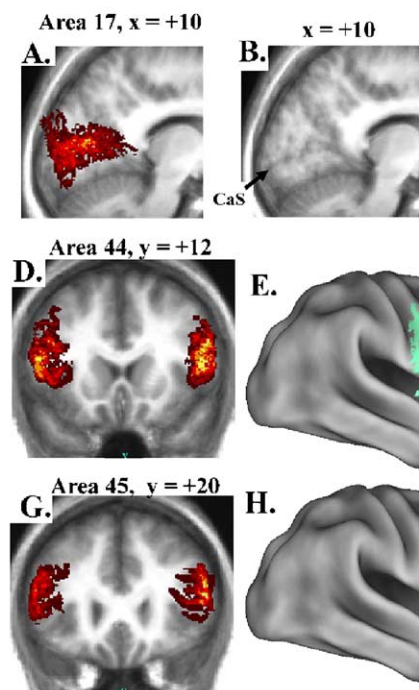
A similar strategy was applied to a set of probabilistic maps of 14 cytoarchitectonic areas identified in a population of individuals and

mapped by volume-based registration to the ‘colin 27’ atlas (Amunts and Zilles, 2001; Amunts et al., 1999, 2000, 2004; Geyer et al., 1996, 1999, 2000; Geyer, 2004; Grekes et al., 2001; Morosan et al., 2001; Rademacher et al., 2001; Zilles et al., 2002). The probabilistic maps for each area [areas 1, 2, 3a, 3b, 4a, 4p, 6, 17 (V1), 18 (V2), A1 (Te 1.0, 1.1, 1.2), 44, and 45] were mapped to the PALS-B12 atlas surface by multi-fiducial mapping (see Methods). Representative results for areas 17, 44, and 45 are shown in Fig. 12. The top row in Fig. 12 shows probabilistic representations of architectonic area 17 from the study of Amunts et al. (2000). Fig. 12A shows a parasagittal slice through the probabilistic volume for area 17, overlaid on the average PALS-B12 structural MRI volume. As expected, area 17 is centered over the calcarine sulcus, which is discernible in the average structural MRI (arrow, Fig. 12B), but its dorso-ventral extent is considerable, in large measure owing to individual variability in the location of the calcarine sulcus after registration. Fig. 12C shows probabilistic architectonic area 17 mapped onto a medial view of the PALS-B12 inflated surface, using a thresholded multi-fiducial map (matched for average surface area) plus an overlay outline of Brodmann’s architectonic area 17 (black contour) from Drury et al. (1999). As expected, the highest probability location for area 17 is in the fundus of the calcarine sulcus; and the overall agreement between these two representations of area 17 is good, given the very different nature of the data and the processing steps. The lower panels of Fig. 12 show corresponding analyses for architectonic areas 44 and 45 (Amunts et al., 1999; 2004), displayed on coronal slices (Figs. 12D, G) and on the right hemisphere inflated map (Figs. 12E, H) and flat map (Figs. 12F, I) with the Drury et al. (1999) version of Brodmann’s areas superimposed on the surface

maps. The agreement between the probabilistic architectonic maps and the Brodmann areas is very good in all three cases illustrated. However, for several other architectonic areas in this data set, the spread of the probabilistic map into inappropriate neighboring sulci is considerably greater (e.g., somatosensory area 3b spreading into the postcentral sulcus). This artifactual spread in the population average is correlated with the variability in 3D location and degree of overlap between nearby sulci in the target atlas volume (cf. Fig. 8).

The high degree of residual individual variability in areal location after volume-based registration was quantified by computing a volume alignment consistency (VAC) index for each architectonic area, just as was done for identified sulci in an earlier section (cf. Evaluation of registration methods section; Table 1). The VAC values ranged from a minimum of 0.05 for Te 1.2 to a maximum of 0.17 for V1 and area 6, with an average of 0.10 for the 14 areas analyzed. These values were lower than the VAC values of 0.11–0.37 for individual sulci, in large part because the individual architectonic volume data consisted of thinner ribbons (~1–2 mm in most regions). After high-dimensional nonlinear registration to the colin atlas (Schormann and Zilles, 1998), the quality of alignment for the same architectonic areas is considerably better as judged by visual inspection (data not shown). However, if the objective is to compare the estimated location of a given architectonic area to other experimental data, such as a population-averaged fMRI activation, then the comparison should be made using data sets registered by equivalent methods (to the degree that this is feasible, given the differences between postmortem vs. in vivo MRI data). Because the great majority of fMRI studies utilize linear or low-dimensional

Probabilistic volumes



Multi-fiducial mapping

Fig. 12. Mapping of probabilistic architectonic areas to the PALS-B12 atlas. Top row: Mapping of area 17 from the probabilistic atlas of Amunts et al. (2004). (A) Parasagittal slice through area 17 in 711-2C space, with the PALS-B12 average structural MRI as an underlay. (B) Average structural MRI from the same parasagittal slice. (C) MFM results displayed on a medial view of the inflated atlas surface; black contour shows area 17 transferred from Brodmann’s original figure to the PALS atlas via Drury et al. (1999). (D) Coronal slice 15 mm anterior to the anterior commissure, through area 44 bilaterally. (E) MFM of area 44 displayed on the average inflated surface. (F) Flat map of area 44. (G) Coronal slice through area 45. (H, I) MFM of area 45 displayed on the inflated map (H) and flat map (I). To view or download these data, and the corresponding left hemisphere data, see <http://sumsdb.wustl.edu:8081/sums/directory.do?dirid=6260387>.

nonlinear registration, the linearly registered probabilistic architectonic volumes are the most appropriate for comparison with most published population-average fMRI data.

Analysis of hemispheric asymmetries

Maps of sulcal depth and sulcal identity that have been registered from individuals to the PALS-B12 atlas provide a valuable substrate for identifying significant group differences that are discernible despite substantial within-group variability. A number of surface-based morphometric (SBM) analysis methods have been developed and are used here to analyze asymmetries between the left and right hemispheres. Hemispheric asymmetries are an interesting topic in their own right (see Discussion) but also serve to illustrate the sensitivity and power of the SBM approach in general.

Surface-based analyses

Fig. 13 compares sulcal depth maps in the left vs. right hemispheres for 24 brains (the PALS-B12 atlas cases plus 12 additional subjects in the same age range). The left column shows the average sulcal depth for the left hemisphere (Fig. 13A) and right hemisphere (Fig. 13D), displayed on lateral views of the highly inflated atlas surface. The STS (black arrows) appears distinctly deeper (darker) on the right than the left hemisphere. The magnitude of this difference is illustrated in a difference map (Fig. 13G), displayed on the right hemisphere surface, with yellow–orange–red regions representing deeper on the right and green–blue regions representing deeper on the left. Much of the STS (centered around $[24^\circ, -128^\circ]_{\text{PALS-B12}}$) is 3–9 mm deeper on the right. In the Sylvian fissure, near the planum temporale (white

arrows in panels A and D), the average depth is 4–7 mm greater on the left than the right (blue in the difference map, centered at $[46^\circ, -146^\circ]_{\text{PALS-B12}}$).

Corresponding views of the medial aspect of the hemisphere (second column, Figs. 13B, E, H) reveal no tightly localized left–right differences, but the overall greenish hue suggests that cingulate cortex may be deeper on average on the left side. On the flat map representations (third column, Figs. 13C, F, I) it is evident that the STS and Sylvian regions of interest constitute the strongest candidates for genuine localized mean depth asymmetries.

Three independent methods were used to evaluate the statistical significance of these candidate asymmetries. In one, the cortex was subdivided into 12 geographically defined subregions, as shown by coloring and outlines in Fig. 13J. The mean sulcal depth within each region was computed for each individual left and right hemisphere, and differences in these mean values were tested for significance by randomization analysis with Bonferroni correction for multiple (12) comparisons. The differences were highly significant for two regions: the STS ($P < 10^{-4}$), and the cingulate ($P < 10^{-3}$); it was also significant for the Sylvian fissure ($P < 0.05$).

In a second approach, the group of 24 brains was split into equal subgroups (group A = cases 1–12; group B = cases 13–24). The depth difference maps for the two subgroups (Figs. 13K, M) are similar, though clearly not identical. Group A was used to identify regions of interest (ROIs) based on contiguous regions having a mean depth difference of a specified magnitude and a surface area of at least 3 cm². Red in Fig. 13L shows the one patch in the STS whose average depth greater on the right by at least 5 mm; blue and green show regions with average depth is greater on the left by at least 3 mm. Using Group A to define the

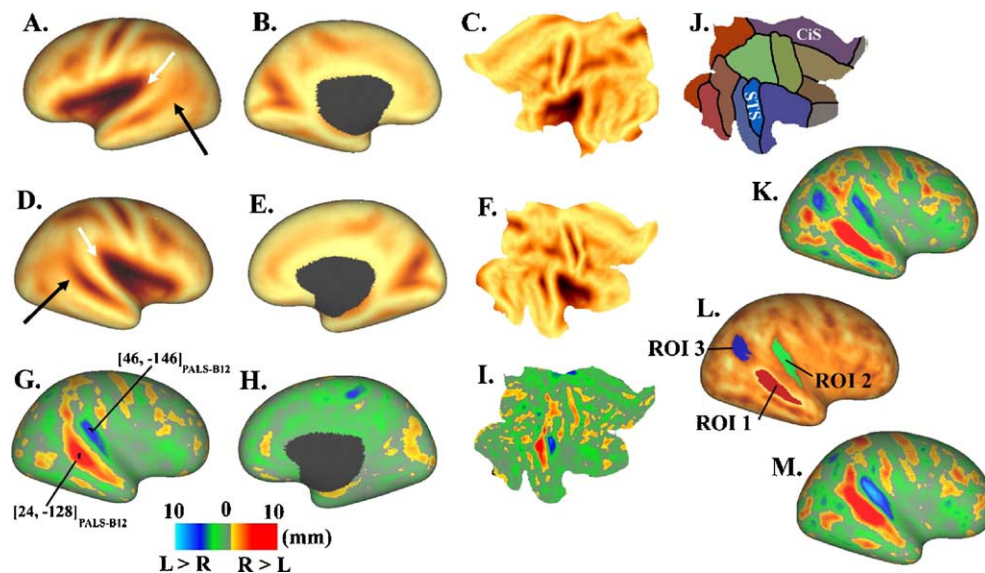


Fig. 13. Left–right asymmetries in sulcal depth maps. (A–C) Population-average sulcal depth map for Cases 1–24 left hemispheres, displayed on lateral (A) and medial (B) views of the average highly inflated PALS-B12 left hemisphere and on the atlas flat map (C). (D–F) Population-average sulcal depth map for Cases 1–24 right hemispheres, displayed on lateral (D) and medial (E) views of the average highly inflated PALS-B12 right hemisphere and on the atlas flat map (F). (G–I) Difference map between the left and right hemisphere average sulcal depth map, with regions deeper on the right shown in yellow and red and regions deeper on the left shown in green and blue, displayed on lateral (G) and medial (H) views of the average highly inflated PALS-B12 right hemisphere and on the atlas flat map (I). (J) Twelve geographic subdivisions of the atlas map used in testing for significance of average depth map differences. The STS and CiS showed significant asymmetries in mean sulcal depth. (K) Difference map for Cases 1–12, the test group, shown only on the lateral view of the highly inflated map because there were no substantial differences on the medial view. (L) Difference map for Cases 1–12 (gold shading) with ROIs 1, 2, 3 shown in red, green, and blue, respectively. (M) Difference map for Cases 13–24, the retest group. Statistical significance for the regional differences in average depth were confirmed for the STS right-hemisphere-deeper region (ROI 1) and the SF left-hemisphere-deeper region (ROI 2) but not for the STS left-hemisphere-deeper region (ROI 3; see Results). Data are available at <http://sumsdb.wustl.edu:8081/sums/directory.do?dirid=6260390>.

ROI and Group B to test its significance, the results show a very significant difference in the retest group for ROI 1 (STS deeper on the right, $P = 10^{-4}$, 1-tailed randomization) and ROI 2 (SF deeper on the left, $P = 10^{-4}$), but not for ROI 3 (dorsal STS deeper on the left, $P = 0.6$).

A third approach involved a cross-correlation analysis to quantify similarities and differences of sulcal depth maps in different individuals in the population of 24 subjects (48 hemispheres). For each of the 2304 (48×48) possible pairwise comparisons, a correlation coefficient was determined using a node-by-node comparison of sulcal depth values. These pairwise correlation coefficients ranged over more than a factor of two (0.31 to 0.65). From visual inspection of the sulcal depth maps, it was evident that pairs with high correlation coefficients (>0.6) appeared relatively similar and that pairs with low correlation coefficients (<0.35) appeared very different in their shape characteristics. Multi-dimensional scaling was applied to the complete correlation matrix, yielding the MDS plot shown in Fig. 14A. The left hemisphere cases (blue) are largely clustered on the left side of the MDS plot, whereas the right hemisphere cases (red) are largely on the right. There is some intermixing, but only three right hemispheres are to the left of the green line and only four left hemispheres are to the right of this divide. The geometric means of these two clusters differ significantly ($P < 10^{-4}$ by randomization test; see Methods). For a finer-grained analysis, the cross-correlation was applied separately to each lobe. The strongest separation was for the temporal lobe, as shown in Fig. 14B. Many of the individual outliers (e.g., 7L, 16L) for the whole hemispheres

are also outliers for the temporal lobe analysis. None of the right hemisphere data points lie to the left of the green line and only five left hemispheres are to the right of this divide ($P < 10^{-5}$, with Bonferroni correction). In contrast, the differences were not significant for the parietal, frontal, or occipital lobes.

The morphological substrate of these temporal lobe shape differences can be examined by viewing the shapes of the STS in individual hemispheres, chosen according to their position on the MDS plot. Fig. 14C shows inflated views of the STS in six hemispheres selected from different regions of the MDS plot (two from the far left, two from the center, and two from the upper right). Fig. 14D (bottom row) shows the sulcal depth maps from the same set of hemispheres after registration to the atlas, displayed on the very inflated right hemisphere surface. Cases 15R and 23R, which are almost superimposed on the temporal lobe MDS plot (gray circle in upper right), have very similar sulcal depth maps in terms of the location and orientation of the STS (green arrows) and pITS (purple arrows). Cases 7L and 11L, near the far left of the MDS plot (ellipse), have sulcal depth maps that differ markedly from Cases 15R and 23R in the vicinity of the STS and pITS. Cases 2L and 3L, near the center of the MDS plot, have sulcal depth maps that are visibly similar to one another and intermediate in characteristics between the other two pairs. In short, position on the MDS plot is reflected in shape characteristics that are discernible in individual hemispheres.

In the MDS plots for the temporal lobe (Fig. 14B), the data points for the left hemisphere (blue) appear to be scattered more widely than for the right hemisphere (red). These differences are statistically significant, based on a comparison of the average

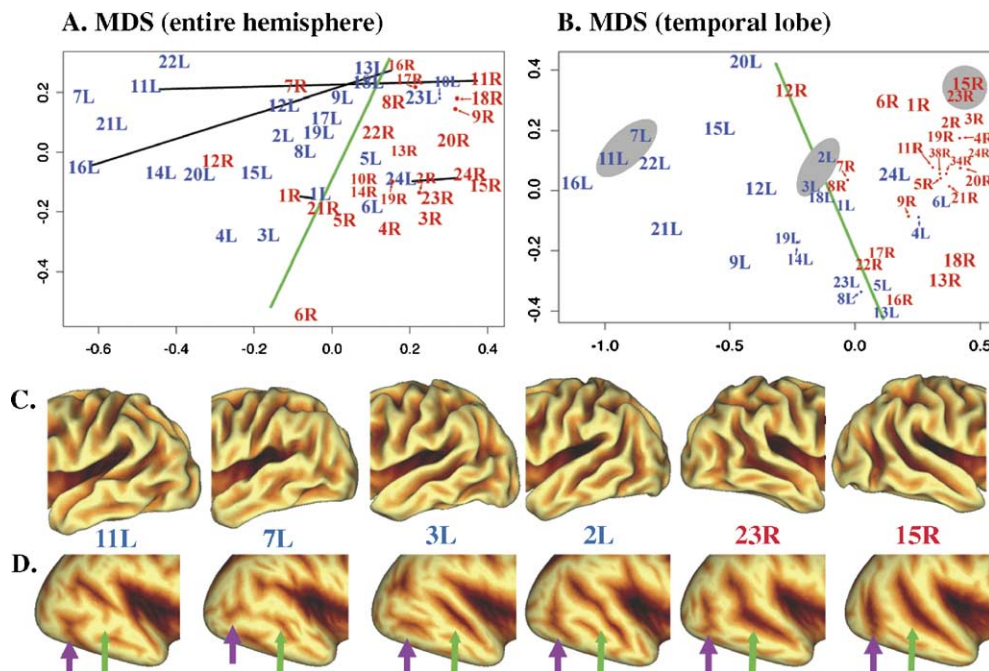


Fig. 14. Multi-dimensional scaling (MDS) analysis of left–right hemispheric asymmetries. (A) MDS results for a matrix of cross-correlation values for all pairwise combinations of 48 hemispheres, with the left hemispheres (1L–24L) shown in blue and the 24 right hemispheres (1R–24R) shown in red. The left hemisphere data points are biased towards the left side of the MDS plot; by a randomization test this is highly significant ($P < 10^{-4}$). Fine lines connect left–right pairs that are smallest (Cases 1, 24) and largest (Cases 11, 16). (B) MDS plot for just the temporal lobe regions of the same 48 hemispheres. (C) Inflated maps of posterior cortex in selected cases, to facilitate assessments of actual shape similarities between cases that are close to one another on the MDS plot in (B) (cases 7L vs. 11L; 3L vs. 2L; 23R vs. 15R) and assessment of shape differences between cases that are far apart on the MDS plot (e.g., 7L or 11L vs. 23R or 15R). (D) Sulcal depth maps for the same six cases displayed on the very inflated PALS-B12 right hemisphere surface. Note that the similarities and differences between cases are easier to discern in this format than on the individual inflated configurations. Data are available at <http://sumsdb.wustl.edu:8081/sums/directory.do?dirid=6260393>.

scatter about the mean, which was 57% greater for the left than the right hemisphere ($P = 0.03$, 2-tailed t test, Bonferroni-corrected). The differences were smaller for the other lobes (range -7% to $+16\%$) and for the entire hemisphere (2%) and did not approach significance for any of them. Thus, by this measure, shape variability is significantly greater for the left than the right hemisphere, but only in temporal cortex.

Another interesting issue is whether the left and right hemispheres of an individual are more similar in shape than would be expected by chance. The black lines in Fig. 14A connect the two left–right pairs that are closest together (Cases 1 and 24) and the two pairs that are farthest apart (Cases 11 and 16). The fact that some pairs are widely spaced implies that hemispheric symmetry is not a dominant characteristic. On the other hand, many of the pairs are in relatively close proximity compared to the mean separation between left and right hemispheres. The average separation of the 24 left–right pairs for the entire hemisphere (0.42 units in Fig. 14A) was 8% smaller than the average separation of all 576 (24×24) possible pairwise left–right combinations of these data (0.45 units), but this difference is not significant ($P = 0.07$, 1-tailed randomization test). Thus, hemispheric symmetry of sulcal depth maps is not significantly greater than would be expected by chance for this group of 24 normal subjects.

Volume-based hemispheric asymmetries

Additional insights regarding left–right asymmetries emerge from a volume-based analysis of the average structural MRI volume and of the probabilistic maps of identified sulci. Panels A–D in Fig. 15 show the average structural MRI in three orthogonal slice planes that intersect a region of high left–right asymmetry near the SF and STS (coronal in panel A, horizontal in panel B, left and right and parasagittal in panels C and D). As is already known

(Toga and Thompson, 2003; Ochiai et al., 2004), the SF (red arrows) and STS (blue arrows) lie more ventral and extend further posterior in the left hemisphere compared to the right in this region. Panels E–G show left–right difference maps for the same slice planes, generated by subtracting the mirror-flipped average sMRI from the original volume. The asymmetry is particularly obvious in the coronal plane, where sulcal gray matter in the left hemisphere is aligned with predominantly white matter in the right hemisphere, and vice versa, resulting in alternating black and white stripes in the difference map. In the parasagittal plane (panel G), the asymmetry is most evident in the dorsal and posterior portions of the STS and SF, near the cross-hairs.

The probabilistic maps of sulcal identity for the 18 largest sulci described in the Probabilistic surface and volume maps of sulcal identity section (Fig. 7) were analyzed for left–right asymmetries. An example is illustrated in Fig. 15 for the STS, shown in coronal (panel H), horizontal (panel I), and parasagittal (left and right sides in panels J and K). Yellow regions show maximal overlap (all 12 maps overlapping in some regions); dark red regions show minimal overlap between individual STS maps. At the levels illustrated, the peaks of the STS maps are slightly (~ 2 – 3 mm) more ventral on the left (panel H) and about 1 cm more posterior (panel I). The asymmetry is very pronounced when viewing the voxel-by-voxel difference between the left and right probabilistic STS maps (panel L). Red–yellow regions represent voxels where the STS is more consistently present in the right hemisphere than in the left; greenish regions represent the converse, where the STS is more prevalent on the left than the right side. The red and green regions are completely segregated in the dorsal and posterior STS. More anteriorly, the differences are smaller and more irregular spatially, signifying a weaker asymmetry in this region.

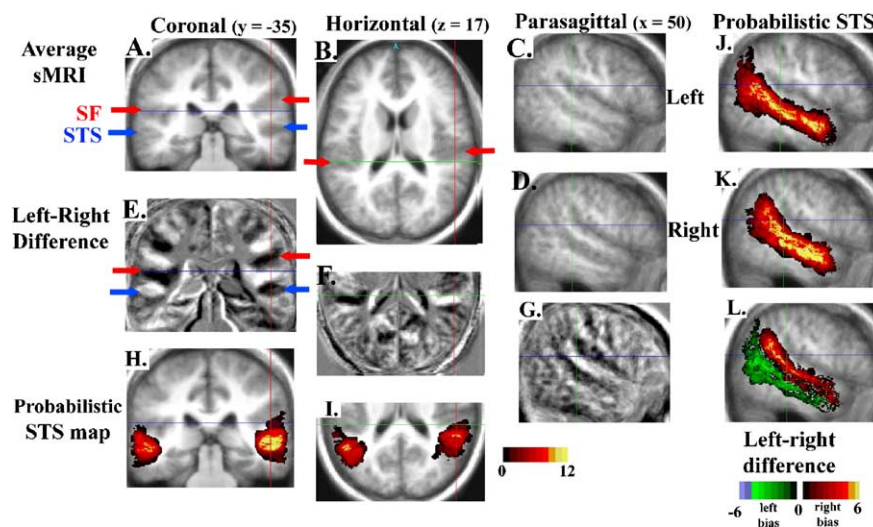


Fig. 15. Hemispheric asymmetries revealed by volume-based analyses. (A–D) Average structural MRI for cases 1–12 viewed in coronal (A), horizontal (B), left parasagittal (C), and right parasagittal (D) slice planes (711–2C space). Arrows centered in the Sylvian Fissure (SF, red) and STS (blue) point to left–right asymmetries. (E–G) Difference maps generated by subtracting the mirror-flipped average sMRI from the original average sMRI, viewed in coronal (E), posterior horizontal (F), and right parasagittal (G) planes. (H–K) Probabilistic map of the STS viewed in coronal (H), posterior horizontal (I), left parasagittal (J) and right parasagittal (K) slice planes. (L) Difference map between the left and right STS viewed in a parasagittal slice. The center of gravity of the population-averaged STS was computed separately for the left hemisphere (stereotaxic coordinates $[-50, -35, +3]_{711-2C}$) and the right hemisphere (coordinates $[+51, -29, +4]_{711-2C}$), i.e., differing by 6 mm along the y axis but only 1 mm along the x and z axes. The statistical significance of these differences was computed for the 10 largest sulci (to retain statistical power after Bonferroni correction). There was a significant left–right asymmetry for the SF (4 mm more posterior on the left, $P < 0.001$), the POS (4 mm more posterior on the left, $P < 0.05$), and the central sulcus (3 mm more posterior on the left, $P < 0.05$), but the STS asymmetry was not significant by this center-of-gravity measure. Data are available at <http://sumsdb.wustl.edu:8081/sums/directory.do?dirid=6276675>.

Fig. 16 shows a more detailed analysis of left–right volumetric asymmetries and their distribution along each major sulcus. Panel A shows the left–right difference for the STS (similar to Fig. 15L but at a more medial slice level) and for the IFS. For this particular slice, the differences appear smaller in magnitude and spatially more irregular for the IFS than for the STS. To evaluate the consistency of this pattern across all slices, the left–right difference maps for the STS and IFS were mapped to the PALS-B12 atlas surface using the multi-fiducial mapping method described above. The results (Fig. 16E) clearly suggest that the asymmetry is indeed more pronounced for the STS than the IFS.

To further quantify the pattern of sulcus-specific asymmetries, the volumetric left–right difference maps were converted from signed values (Fig. 16A) to absolute (unsigned) values. The resultant hemispheric difference magnitude map is shown for the STS and IFS in Fig. 16B and as a composite for the 18 largest sulci in Fig. 16C. The complete pattern across the hemisphere for this composite difference map is shown on the atlas flat map after multi-fiducial mapping (Fig. 16D). The large regional differences in this pattern reflect not only genuine asymmetries but also regional differences in the sampling density of the probabilistic sulcal maps, since the 18 selected sulci do not encompass the entire hemisphere. To compensate for regional sampling differences, the sulcal sampling density was computed separately for each sulcus. This is shown for the STS plus IFS in Fig. 16F (the sum of the left and mirror-flipped right probabilistic maps) and for all 18 sulci in Fig. 16G. This 18-sulcus composite volume was mapped onto the atlas surface and then thresholded to remove regions with low sampling density (Fig. 16H). A normalized estimate of hemispheric sulcal asymmetry was computed by taking the ratio between the

left–right difference map (Fig. 16D) and the thresholded sulcal sampling density map (panel H). The resultant sulcal asymmetry map, shown on the inflated atlas surface in Figs. 16I, J and on the flat map in Fig. 16K, has a peak near the dorsal tip of the STS, at spherical coordinates $[31^\circ, -158^\circ]_{\text{PALS-B12}}$. It is instructive to compare this locus with the previously illustrated sulcal depth difference maps (Figs. 15G, I, K) which showed peak differences in average sulcal depth in the STS at $[24^\circ, -128^\circ]_{\text{PALS-B12}}$ and in the SF at $[46^\circ, -146^\circ]_{\text{PALS-B12}}$ (see blue marks in Figs. 16I, K). The volume-based and surface-based asymmetries differ by about 2 cm, indicating that the two sets of analyses (surface-based sulcal depth and volume-based sulcal identity) provide complementary yet correlated information about hemispheric asymmetries.

Discussion

This study was carried out with six major objectives in mind: (i) to introduce the PALS atlas approach as a general way to cope with the complexity of human cortical convolutions and their variability across individuals; (ii) to make the PALS-B12 atlas data set available, including a wide variety of surface and volume data, along with a flexible set of online and offline visualization tools; (iii) to evaluate and compare the quality of surface-based and volume-based registration; (iv) to introduce new surface-based morphometry (SBM) methods and to demonstrate their utility in characterizing left–right asymmetries in cortical structure; (v) to illustrate a new population-based strategy for mapping group-average fMRI and architectonic data onto atlas surfaces; and (vi) to integrate the PALS-B12 atlas into a database infrastructure that

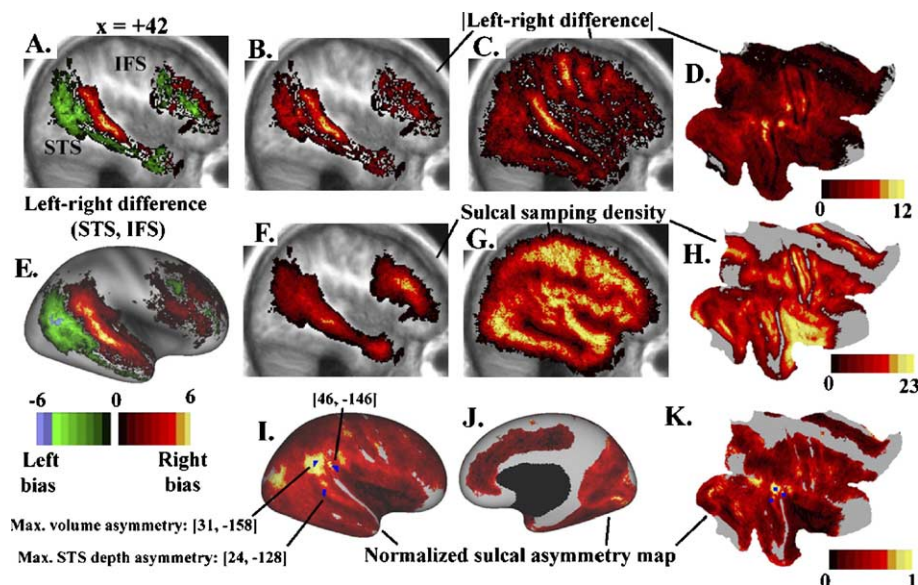


Fig. 16. Mapping of volumetric sulcal asymmetries. (A) Left–right difference map for the STS and IFS (parasagittal slice 7 mm more medial than in Fig. 15L). (B) Difference magnitude map (absolute value of the left–right difference) for the slice shown in (A). (C) Difference magnitude map for all 18 sulci listed in Table 1. (D) Difference magnitude map displayed on the atlas flat map after multi-fiducial mapping. (E) Left–right difference map displayed on the very inflated PALS-B12 surface after multi-fiducial mapping from the volume illustrated in (A). (F) Sulcal sampling density map for the STS and IFS, computed by summing the left and right probabilistic maps of each hemisphere, shown in the same slice as in (A–C). (G) Sulcal sampling density for all 18 sulci. (H) Sulcal sampling density displayed on the atlas flat map after multi-fiducial mapping followed by thresholding (at a level of 3.0) to eliminate regions with low sampling density. (I–K) Normalized sulcal asymmetry map, generated by dividing the left–right difference magnitude map (D) by the difference magnitude (H) and setting the value at 0 for all below-threshold values on the sulcal density map. Green marks in the very inflated map in (I) indicate centers of maximal STS and SF sulcal depth differences in Fig. 15G. Data are available at <http://sumsdb.wustl.edu:8081/sums/directory.do?dirid=6276678>.

allows ready access to existing data sets and easy incorporation of additional data sets. Each of these topics warrants brief discussion.

Principles underlying the PALS approach

The essence of the PALS atlas approach is to register individual hemisphere surfaces to an atlas sphere, using selected geographic landmarks on each individual and a population average of these landmarks as the target. The rationale underlying this strategy is based on several organizational and developmental principles that relate functional organization to the pattern of cortical convolutions. (i) *The cortical mosaic.* Human cerebral cortex contains a complex mosaic of anatomically and functionally distinct areas, perhaps 100–200 in total number, ranging more than 10-fold in average surface area (Van Essen, 2004b; Öngür et al., 2003). These are interconnected by thousands of cortico-cortical pathways, if monkey studies are a reasonable guide (Felleman and Van Essen, 1991). Across individuals, there is twofold or more variability in the size of each cortical area (Stensaas et al., 1974; Dougherty et al., 2003) and presumably also in the strength of various connections (pathways). (ii) *Areal boundaries and convolutions.* The correlation between areal boundaries and sulcal/gyral folds is strongest for primary sensory–motor areas and primary cortical sulci, but over these regions the correspondence varies from 3 to 5 mm for portions of areas 3b and A1 (Te 1.0) to 1–2 cm for areas 17 and 18 (Rademacher et al., 1993; Amunts et al., 2000; Morosan et al., 2001; Dougherty et al., 2003). In most other regions, the correlations are weaker (Amunts et al., 1999, 2000; Öngür et al., 2003). This contrasts with macaque cortex, where areal boundaries are consistently correlated with sulcal/gyral boundaries to within ~2–3 mm over most of the hemisphere (Lewis and Van Essen, 2000, Van Essen, 2004b). (iii) *Developmental mechanism for cortical folding.* The pattern of cortical convolutions and their species-specific relationships with areal boundaries can be explained by the hypothesis that mechanical tension along long-distance cortico-cortical pathways is the primary driving force for cortical folding (Van Essen, 1997). By this hypothesis, consistent folding patterns occur in regions where connectivity is dominated by relatively few pathways between large areas, whereas folding patterns are less consistent and less well correlated with areal boundaries in ‘balkanized’ regions containing a mosaic of smaller areas whose interconnections are more complex and variable.

These relationships between cortical shape and function impose fundamental constraints on what can be achieved when registering individuals to an atlas. It is not possible for geographic and functional landmarks to be simultaneously brought into perfect alignment. Moreover, just considering geography alone, perfect registration of geographic landmarks throughout the hemisphere is a flawed concept, because precisely corresponding geographic landmarks cannot be reliably identified in regions of high shape variability (cf. Fig. 6N). Considering just function alone, perfect registration of functionally defined areas is also inherently problematic, given the nature of individual variability in areal size. Even if a set of well-defined areas (e.g., primary sensory areas) were brought into precise register, the registration in intervening regions would be far from perfect, owing to individual variability in the relative sizes of the many intervening areas.

Evaluating registration quality

The above considerations highlight the need for quantitative evaluations of how well different registration methods reduce various types of variability. The present study included analyses of spatial distortions at successive stages of surface-based analysis and of the consistency with which identified sulci are brought into register using SBR and VBR. Spatial distortions are inevitable, given the pattern of intrinsic curvature on the cortex, just as distortions would be inevitable when inflating or flattening a hypothetical rubber glove that contains many local bumps and dimples. Analysis of these distortions supports several general observations regarding the PALS-B12 atlas data set. Areal distortions arise mainly during inflation and mapping to the individual standard spherical map; distortions in the spherical registration stage are generally small and vary smoothly. Most importantly, the general pattern of spatial distortions is consistent across individuals, which reduces the negative impact of distortions on registration quality.

Registration metrics using identified sulci as substrates demonstrated that SBR using the Caret/PALS approach achieves quantitatively good performance across the entire hemisphere, based on high consistency in the alignment of each sulcus and low overlap between adjacent sulci. In contrast, VBR using both an affine transformation and a low-dimensional nonlinear transformation resulted in markedly lower consistency of sulcal alignment and greater overlap between adjacent sulci. Fundamentally, this is because low-dimensional VBR algorithms have insufficient degrees of freedom to produce the complex displacement fields needed to achieve good sulcal alignment in the volume. A very high-dimensional VBR can in principle perform better (e.g., Schormann and Zilles, 1998), but unless the algorithm effectively invokes surface-related constraints as in SBR, there is a risk that sulcal/gyral features in the individual may be forced into register with atlas features that are in geographic proximity but do not reflect functional correspondences (e.g., if a gyral region in one individual is forced to match to a gyral region on the atlas even though the functionally corresponding region may lie in a sulcus in the atlas). A related concern is that the nonlinearities in a high-dimensional registration algorithm can lead to large, local irregularities in the deformation field, resulting in biologically unnatural distortions (compression, expansion, shear).

Similar issues arise in comparing different SBR algorithms. The SBR algorithms provided in FreeSurfer (Fischl et al., 1999b) and BrainVoyager (<http://www.brainvoyager.com>) software differ from the Caret/PALS algorithm in ways that impact registration quality. For example, FreeSurfer SBR aims to achieve a good fit between continuously varying maps of average convexity (similar to sulcal depth) for the individual and the population-average target map (Fischl et al., 1999b). This energy minimization approach can achieve excellent alignment of structures that have similar shape characteristics. However, in regions of high shape variability, the algorithm can force alignment between regions that do not necessarily correspond either geographically or functionally. As just discussed for volume registration, this can result in large local distortions that are unlikely to reflect actual functional variability. In view of such considerations, the relative advantages and limitations of different SBR and high-dimensional VBR approaches need extensive evaluation and quantitative comparison.

Does the superior performance of Caret/PALS SBR over VBR in bringing identified sulci into alignment also extend to analyses

of fMRI activation patterns in a population of subjects? This is likely to be the case, given the known correlations between cortical function and sulcal patterns. However, it will be important to test this prediction quantitatively, using metrics similar to those presented here, but designed to assess the consistency of alignment of fMRI activation foci in subjects tested with a particular behavioral paradigm. Future refinements in SBR algorithms or in selection of appropriate registration landmarks will presumably lead to further improvements, but the current PALS-B12 atlas and registration strategy provides a new and well-defined standard against which to compare other methods.

Coordinate systems, atlas transformations, and point-to-cloud mappings

The PALS-B12 atlas includes a primary surface-based coordinate system, based on latitude and longitude on the atlas sphere. In addition, it is linked to six different stereotaxic coordinate systems and to two surface-based atlases in common use. Atlas coordinates allow precise specification of location in a particular atlas space, but the mappings from one coordinate system to another are inherently more complex. The PALS-B12 surface can be linked to other surface-based atlases by point-to-point mappings. However, any number of mappings are possible, depending on the registration constraints and algorithm used, and it is not straightforward to evaluate the fidelity with which any one mapping brings biologically corresponding locations into register. For the mapping between the PALS-B12 surface and stereotaxic volumes the relationship is even more complex because each mapping is point-to-cluster (point-to-cloud) rather than one-to-one. Despite its complexity, this is a more realistic and objective way to address the inherent uncertainty of mappings between atlases (see also Fig. 12 in MacDonald et al., 2000).

Experimental data on the atlas

A general objective is to make the PALS-B12 atlas (or its successors) a useful repository for a rapidly growing amount of experimental data originating from many different laboratories. Many issues must be addressed for this objective to be attained, given the complexities of the experimental data, the diverse formats in which they can be stored and displayed, and the diversity of ways that data can be mapped onto the atlas (directly or via other atlas surfaces or volumes). One issue is that the processing steps for mapping data to the atlas should be as flexible and easy to carry out as possible. To this end, Caret provides a variety of methods for mapping data onto the PALS atlas. These include mapping from an individual surface to the atlas by SBR (flat map and spherical map); volume-to-surface mapping by MFM or by mapping to the average fiducial surfaces; and projection of stereotaxic foci. For most of these options, the requisite steps are explained in tutorials that are available online.

Another important need is to include key metadata associated with each data set. Experimental data can be useless or misleading if information is not available regarding the source and nature of the data and critical processing steps. A general strategy in implementing the PALS-B12 atlas has been to insure that each data set is accompanied by key metadata that reference the original source(s) of experimental data and track the processing steps involved in mapping to the atlas. Most Caret file formats for spatial data include comment options that allow open-ended inclusion of

relevant metadata, and many of the processing steps are automatically included. Additional issues of databases and data management are discussed in a later section.

Multi-fiducial mapping

Volume-averaged group data are widely used in fMRI studies, mainly because the approach increases statistical power. There are obvious advantages to mapping volume-averaged data onto an atlas surface to aid in visualization. MFM provides an effective solution to the longstanding problem of biases from mapping to the idiosyncratic shape of any given individual atlas surface. This results in improved estimates of the most likely activation pattern and of the total range of uncertainty. The use of 12 surfaces for the current MFM implementation represents an effective compromise, given the computation time and data storage needs that a substantially larger number would entail. To make MFM a widely applicable approach, Caret includes an option for batch processing of group-average fMRI data to any of six commonly used stereotaxic spaces.

In order to interpret the results of MFM, it is important to consider several underlying assumptions. Without access to the individual structural and fMRI data in any given study, it is impossible to work backwards from volume-averaged group data to determine what the actual pattern would be in any individual. Hence, the activations seen on any of the individual PALS-B12 surfaces do not reflect a pattern in fact attributable to any of the actual fMRI subjects. Nor do they necessarily reflect the pattern that would have arisen in the 12 subjects whose structural data contributed to the atlas if they had been tested using the same fMRI paradigm. MFM does provide an objective strategy for estimating both the region of most likely activation and a plausible upper bound on the total extent of activation. This constitutes an important advance over the common current practice of mapping volume-averaged results onto a single-subject atlas.

In many situations, it is appropriate to map group-average data using both AFM and MFM. The two mapping methods yield similar but not identical spatial patterns and are inherently complementary. AFM is conceptually simpler and allows readout of values at each surface node that correspond to a particular voxel value. MFM provides a more objective assessment of the most likely spatial distribution on the atlas surface.

Hemispheric asymmetries and surface-based morphometry

Left–right asymmetries in cortical structure have been analyzed using a variety of methods that are sensitive to different aspects of cortical morphology. Quantitative approaches applicable to structural MRI data include ROI methods that compare manually identified regions of interest (e.g., Loftus et al. 1993; Ochiai et al., 2004); voxel-based morphometry (VBM) methods that compare the distribution of voxels categorized as gray matter, white matter, and ventricular (e.g., Good et al., 2001; Watkins et al., 2001); and surface-based methods that compare the distribution of sulcal contours along the external hull of the cerebrum (Thompson et al., 2000, 2001; Sowell et al., 2002). As summarized by Toga and Thompson (2003), these analyses have provided improved assessments of various asymmetries described in earlier postmortem studies, including frontal petalia (right protruding), occipital petalia (left protruding), and temporal lobe asymmetries in the perisylvian language region, including the

planum temporale of the SF (larger on the left) and the dorsal STS and SF (more anterior on the right).

The SBM analyses introduced here are complementary to the methods noted above for analyzing shape differences between groups, and they show greater sensitivity to several aspects of cortical shape. For example, VBM does not incorporate information about surface topology and hence is not well suited for analyzing subtle shape asymmetries related to specific sulci or folding patterns. The surface-hull approach (Thompson et al., 2000, 2001; Sowell et al., 2002) maps features on and near the external hull of the hemisphere but is insensitive to shape characteristics in deeply buried sulcal regions. In general, the cortical asymmetries reported here are consistent with the patterns described in previous reports, but they add valuable additional information. One advance involves accurate quantification and spatial localization of shape characteristics. For example, consider the findings reported here that the STS is deeper on the right side by 5 mm or more on average in a region centered on $[24^\circ, -128^\circ]_{\text{PALS-B12}}$ and that the SF is deeper on the left by 3 mm or more on average in a region centered on $[46^\circ, -146^\circ]_{\text{PALS-B12}}$. These findings are consistent with the asymmetries reported by Ochiai et al. (2004) in a volume ROI analysis of the STS, but they provide greater spatial detail and precision. Another advance involves quantitative analyses of shape similarities and differences at the level of individual subjects, particularly using multi-dimensional scaling (Fig. 12). This includes the ability to identify individuals who have relatively unusual shape characteristics (outliers on the MDS plot) versus those who are more typical insofar as they are near the center of the MDS plot. The finding that shape variability is significantly greater for the left hemisphere in the temporal lobe but not in other lobes fits with evidence that the length of the STS is more variable in the left hemisphere (Ochiai et al., 2004) but extends the analysis to other shape characteristics. A third advance involves the quantification of volumetric asymmetries, which provides additional information about hemispheric asymmetries within the STS and other identified sulci.

More generally, SBM can be used for a wide range of comparisons across groups that differ in clinical diagnosis. For example, this approach has revealed specific shape differences in parietal cortex for Williams Syndrome individuals vs. age-matched normal controls (Van Essen et al., 2004) and also shape differences in autistic children vs. age-matched normals (Nordahl et al., *in press*). These differences have been revealed using relatively small sample sizes (e.g., 24 normal individuals in the present study). Of course, when greater numbers of subjects are available, this will increase the sensitivity to subtle differences.

Options for target landmarks

An important methodological issue concerns the choice of target landmarks for various types of analysis, particularly for morphometric comparisons between different groups. The PALS-B12 surface data sets illustrated in this study were all registered to target landmarks averaged across left and right hemispheres of a population of right-handed young adult males and females, thus circumventing biases that might arise from choosing just the left or the right side as the registration target. While this is appropriate for analyses of hemispheric asymmetries in normal adults (cf. Figs. 15 and 16), there are other group comparisons in which alternative targets for the registration process warrant consider-

ation. Fortunately, it is relatively straightforward to generate and use these alternative landmark targets. For example, in a study of gender differences in cortical shape, data for the left and right hemispheres were analyzed separately and were registered to hemisphere-specific targets (e.g., left hemispheres registered to a population-average target derived only from left hemisphere landmarks) as well as to the hybrid left and right landmarks used in the present study. The results for each group were only marginally affected when comparing results for hemisphere-specific vs. hybrid left–right targets. There were only minor differences in the precise locations (coordinates) on the target map of various features in the difference maps between groups and in the exact pattern of data points for individual hemispheres in the MDS plots.

Database and access issues

The complete PALS-B12 atlas data set is diverse and complex, as it includes surface and volume data for population averages as well as individuals, representations of atlas surfaces and volumes in many stereotaxic spaces, and many types of ancillary data. The total amount of publicly available atlas-related data in Sums DB, is much greater than any individual investigator will generally wish to download. Hence, it has been subdivided into figure-specific components that can all be visualized online (via WebCaret), downloading can be limited to particular subsets needed for offline visualization and analysis.

With a growing amount of experimental data mapped to the PALS-B12 atlas and made freely accessible in Sums DB, there will be a concomitant need for more powerful and flexible options for searching data within and across databases. The ability to easily interrogate Sums DB using spatially based queries and text queries that reflect structured neurobiological terminologies (e.g., different cortical areas, different partitioning schemes, different behavioral tasks in fMRI studies) will open new vistas for data mining. These and related advances on the neuroinformatics front hold prospects for dramatic improvements in web-based sharing and communication of structural and functional neuroimaging data by the neuroscience community.

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References

- Amunts, K., Zilles, K., 2001. Advances in cytoarchitectonic mapping of the human cerebral cortex. In: Naidich, T.P., Yousry, T.A., Mathews, V.P. (Eds.), *Neuroimaging Clinics of North America. Anatomic Basis of Functional MR Imaging*. Harcourt, Philadelphia, pp. 151–169.

- Amunts, K., Schleicher, A., Burgel, U., Mohlberg, H., Uylings, H.B., Zilles, K., 1999. Broca's region revisited: cytoarchitecture and intersubject variability. *J. Comp. Neurol.* 412, 319–341.
- Amunts, K., Malikovic, A., Mohlberg, H., Schormann, T., Zilles, K., 2000. Brodmann's areas 17 and 18 brought into stereotaxic space—Where and how variable? *NeuroImage* 11, 66–84.
- Amunts, K., Weiss, P.H., Mohlberg, H., Pieperhoff, P., Eickhoff, S., Gurd, J.M., Marshall, J.C., Shah, N.J., Fing, G.R., Zilles, K., 2004. Analysis of neural mechanisms underlying verbal fluency in cytoarchitectonically defined stereotaxic space—The roles of Brodmann areas 44 and 45. *NeuroImage* 22, 42–56.
- Brodman, K., 1909. Vergleichende Lokalisationslehre der Grosshirnrinde in ihren Prinzipien dargestellt auf Grund des Zellenbaues. J.A. Barth, Leipzig.
- Buckner, R.L., Raichle, M.E., Petersen, S.E., 1995. Dissociation of human prefrontal cortical areas across different speech production tasks and gender groups. *J. Neurophysiol.* 74, 2163–2173.
- Buckner, R.L., Head, D., Parker, J., Fotenos, A.F., Marcus, D., Morris, J.C., Snyder, A.Z., 2004. A unified approach for morphometric and functional data analysis in young, old, and demented adults using automated atlas-based head size normalization: reliability and validation against manual measurement of total intracranial volume. *Neuroimage* 23, 724–738.
- Chung, M.K., Worsley, K.J., Robbins, S., Paus, T., Taylor, J., Giedd, J.N., Rapoport, J.L., Evans, A.C., 2003. Deformation-based surface morphometry applied to gray matter deformation. *NeuroImage* 18, 198–213.
- Corbetta, M., Shulman, G.L., 2002. Control of goal-directed and stimulus-driven attention in the brain. *Nat. Rev. Neurosci.* 3, 201–215.
- Corbetta, M., Kincade, J.M., Ollinger, J.M., McAvoy, M.P., Shulman, G.L., 2000. Voluntary orienting is dissociated from target detection in human posterior parietal cortex. *Nat. Neurosci.* 3, 292–297 (Erratum in: *Nat. Neurosci.* 2000 May 3(5:521)).
- Desai, R., Liebenthal, E., Possing, E.T., Waldron, E., Binder, J.R., 2005. Volumetric vs. surface-based alignment for localization of auditory cortex activation. *Neuroimage* 16, 1019–1029.
- Dougherty, R.F., Koch, V.M., Brewer, A.A., Fischer, B., Modersitzki, J., Wandell, B.A., 2003. Visual field representations and locations of visual areas V1/2/3 in human visual cortex. *J. Vis.* 3, 586–598.
- Drury, H.A., Van Essen, D.C., Anderson, C.H., Lee, C.W., Coogan, T.A., Lewis, J.W., 1996. Computerized mappings of the cerebral cortex. A multiresolution flattening method and a surface-based coordinate system. *J. Cogn. Neurosci.* 8, 1–28.
- Drury, H.A., Van Essen, D.C., Corbetta, M., Snyder, A.Z., 1999. Surface-based analyses of the human cerebral cortex. In: Toga, A., et al., (Eds.), *Brain Warping*. Academic Press, pp. 337–363.
- Evans, A.C., Collins, D.I., Neelin, P., 1994. Three-dimensional correlative imaging: applications in human brain mapping. In: Thatcher, R.W., Hallett, M., Zeffiro, T., et al., (Eds.), *Functional Neuroimaging: Technical Foundations*. Academic Press, New York, pp. 145–162.
- Felleman, D.J., Van Essen, D.C., 1991. Distributed hierarchical processing in primate cerebral cortex. *Cereb. Cortex* 1, 1–47.
- Fischl, B., Sereno, M.I., Dale, A.M., 1999. Cortical surface-based analysis: II. inflation, flattening, and a surface-based coordinate system. *NeuroImage* 9, 195–207.
- Fischl, B., Sereno, M.I., Tootell, R.B., Dale, A.M., 1999. High-resolution intersubject averaging and a coordinate system for the cortical surface. *Hum. Brain Mapp.* 8, 272–284.
- Friston, K.J., Holmes, A.P., Worsley, K.J., Poline, J.P., Frith, C.D., Frackowiak, R.S.J., 1995. Statistical parametric maps in functional imaging: a general linear approach. *Hum. Brain Mapp.* 2, 189–210.
- Geyer, S., 2004. The microstructural border between the motor and the cognitive domain in the human cerebral cortex. In: Beck, F., Christ, B., Kriz, W., Marani, E., Kummer, W., Putz, R., Sano, Y., Schiebler, T.H., Zilles, K. (Eds.), *Advances in Anatomy, Embryology and Cell Biology*. Springer, Berlin.
- Geyer, S., Ledberg, A., Schleicher, A., Kinomura, S., Schormann, T., Burgel, U., Klingberg, T., Larsson, J., Zilles, K., Roland, P.E., 1996. Two different areas within the primary motor cortex of man. *Nature* 382, 805–807.
- Geyer, S., Schleicher, A., Zilles, K., 1999. Areas 3a, 3b, and 1 of human primary somatosensory cortex. *Neuroimage* 10, 63–83.
- Geyer, S., Schormann, T., Mohlberg, H., Zilles, K., 2000. Areas 3a, 3b, and 1 of human primary somatosensory cortex: Part 2. Spatial normalization to standard anatomical space. *Neuroimage* 11, 684–696.
- Geyer, S., Schleicher, A., Schormann, T., Mohlberg, H., Bodegard, A., Roland, P.E., Zilles, K., 2001. Integration of microstructural and functional aspects of human somatosensory areas 3a, 3b, and 1 on the basis of a computerized brain atlas. *Anat. Embryol.* 204, 351–366.
- Good, C.D., Johnsrude, I., Ashburner, J., Henson, R.N., Friston, K.J., Frackowiak, R.S., 2001. Cerebral asymmetry and the effects of sex and handedness on brain structure: a voxel-based morphometric analysis of 465 normal adult human brains. *Neuroimage* 14, 685–700.
- Grefkes, C., Geyer, S., Schormann, T., Roland, P., Zilles, K., 2001. Human somatosensory area 2: observer-independent cytoarchitectonic mapping, interindividual variability, and population map. *Neuroimage* 14, 617–631.
- Hadjikhani, N., Liu, A.K., Dale, A.M., Cavanagh, P., Tootell, R.B.H., 1998. Retinotopy and color sensitivity in human visual cortical area V8. *Nat. Neurosci.* 1, 235–241.
- Head, D., Snyder, A.Z., Gorton, L.E., Morris, J.C., Buckner, R.L., in press. Frontal–hippocampal double dissociation between normal aging and Alzheimer's disease. *Cereb. Cortex* (Sep 15).
- Lewis, J.W., Van Essen, D.C., 2000. Mapping of architectonic subdivisions in the macaque monkey, with emphasis on parieto-occipital cortex. *J. Comp. Neurol.* 428, 79–111.
- Loftus, W.C., Tramo, M.J., Thomas, C.E., Green, R.L., Nordgren, R.A., Gazzaniga, M.S., 1993. Three-dimensional quantitative analysis of hemispheric asymmetry in the human superior temporal region. *Cereb. Cortex* 3, 348–355.
- Lohman, G.D., von Cramon, D.Y., Steinmetz, H., 1999. Sulcal variability of twins. *Cereb. Cortex* 9, 754–763.
- MacDonald, D., Kabani, N., Avis, D., Evans, A.C., 2000. Automated 3-D extraction of inner and outer surfaces of cerebral cortex from MRI. *NeuroImage* 12, 340–356.
- Mazziotta, J., Toga, A., Evans, A., Fox, P., Lancaster, J., Zilles, K., Woods, R., Paus, T., Simpson, G., Pike, B., Holmes, C., Collins, L., Thompson, P., MacDonald, D., Iacoboni, M., Schormann, T., Amunts, K., Palomero-Gallagher, N., Geyer, S., Parsons, L., Narr, K., Kabani, N., Le Goualher, G., Boomsma, D., Cannon, T., Kawashima, R., Mazoyer, B., 2001. A probabilistic atlas and reference system for the human brain: International Consortium for Brain Mapping (ICBM). *Philos. Trans. R Soc. Lond., B Biol. Sci.* 356, 1293–1322.
- Morosan, P., Rademacher, J., Schleicher, A., Amunts, K., Schormann, T., Zilles, K., 2001. Human primary auditory cortex: cytoarchitectonic subdivisions and mapping into a spatial reference system. *Neuroimage* 13, 684–701.
- Nordahl, C.W., Mostafavi, I., Hanlon, D., Amaral, D.G., VanEssen, D.C., in press. Cortical shape analyses in low functioning autism. *International Meeting for Autism Research*. Boston, MA.
- Ochiai, T., Grimault, S., Scavarda, D., Roch, G., Hori, T., Riviere, D., Mangin, J.F., Regis, J., 2004. Sulcal pattern and morphology of the superior temporal sulcus. *Neuroimage* 22, 706–719.
- Ojemann, J.G., Akbudak, E., Snyder, A.Z., Mckinstry, R.C., Raichle, M.E., Conturo, T.E., 1997. Anatomic localization and quantitative analysis of gradient refocused echo-lanaar fMRI susceptibility artifacts. *Neuroimage* 6, 156–167.
- Öngur, D., Ferry, A.T., Price, J.L., 2003. Architectonic subdivision of the human orbital and medial prefrontal cortex. *J. Comp. Neurol.* 460, 425–449.
- Ono, M., Kubick, S., Abernathy, C.D., 1990. *Atlas of the Cerebral Sulci*. Thieme Medical, New York.
- Rademacher, J., Caviness Jr., V.S., Steinmetz, H., Galaburda, A.M., 1993. Topographical variation of the human primary cortices: implications for

- neuroimaging, brain mapping, and neurobiology. *Cereb. Cortex* 3, 313–329.
- Rademacher, J., Morosan, P., Schormann, T., Schleicher, A., Werner, C., Freund, H.J., Zilles, K., 2001. Probabilistic mapping and volume measurement of human primary auditory cortex. *Neuroimage* 13, 669–683.
- Riviere, D., Mangin, J.F., Papadopoulos-Orfanos, D., Martinez, J.M., Frouin, V., Regis, J., 2002. Automatic recognition of cortical sulci of the human brain using a congregation of neural networks. *Med. Image Anal.* 6, 77–92.
- Roland, P.E., Graufelds, C.J., Wahlin, J., Ingelman, L., Andersson, M., Ledberg, A., Pedersen, J., Akerman, S., Dabringhaus, A., Zilles, K., 1994. Human brain atlas for high-resolution functional and anatomical mapping. *Hum. Brain Mapp.* 1, 173–184.
- Saad, Z.S., Reynolds, R.C., Argall, B., Japee, S., Cox, R.W., 2004. UMA: an interface for surface-based intra- and inter-subject analysis with AFNI. Proceedings of the 2004 IEEE International Symposium on Biomedical Imaging, pp. 1510–1513.
- Salat, D.H., Buckner, R.L., Snyder, A.Z., Greve, D.N., Desikan, R.S., Busa, E., Morris, J.C., Dale, A.M., Fischl, B., 2004. Thinning of the cerebral cortex in aging. *Cereb. Cortex* 14, 721–730 (electronic publication 2004 Mar 28).
- Schormann, T., Zilles, K., 1998. Three-dimensional linear and nonlinear transformations: an integration of light microscopical and MRI data. *Hum. Brain Mapp.* 6, 339–347.
- Sowell, E.R., Thompson, P.M., Rex, D., Kornsand, D., Tessner, K.D., Jernigan, T.L., Toga, A.W., 2002. Mapping sulcal pattern asymmetry and local cortical surface gray matter distribution in vivo: maturation in perisylvian cortices. *Cereb. Cortex* 12, 17–26.
- Stensaas, S.S., Eddington, D.K., Dobelle, W.H., 1974. The topography and variability of the primary visual cortex in man. *J. Neurosurg.* 40, 747–755.
- Talairach, J., Tournoux, P., 1988. *Coplanar Stereotaxic Atlas of the Human Brain*. Thieme Medical, New York.
- Thompson, P.M., Woods, R.P., Mega, M.S., Toga, A.W., 2000. Mathematical/computational challenges in creating deformable and probabilistic atlases of the human brain. *Hum. Brain Mapp.* 9, 81–92.
- Thompson, P.M., Mega, M.S., Woods, R.P., Zoumalan, C.I., Lindshield, C.J., Blanton, R.E., Moussai, J., Holmes, C.J., Cummings, J.L., Toga, A.W., 2001. Cortical change in Alzheimer's disease detected with a disease-specific population-based brain atlas. *Cereb. Cortex* 11, 1–16.
- Toga, A.W., 1999. *Brain Warping*. Academic Press (385 pp.).
- Toga, A.W., Thompson, P.M., 2002. New approaches in brain morphometry. *Am. J. Geriatr. Psychiatry* 10, 13–23.
- Toga, A.W., Thompson, P.M., 2003. Mapping brain asymmetry. *Nat. Rev. Neurosci.* 4, 37–48.
- Van Essen, D.C., 1997. A tension-based theory of morphogenesis and compact wiring in the central nervous system. *Nature* 385, 313–318.
- Van Essen, D.C., 2002. Windows on the brain. The emerging role of atlases and databases in neuroscience. *Curr. Opin. Neurobiol.* 12, 574–579.
- Van Essen, D.C., 2004a. Surface-based approaches to spatial localization and registration in primate cerebral cortex. *NeuroImage* 23, s97–s107 (Supplement. Mathematics in Brain Imaging).
- Van Essen, D.C., 2004b. Organization of visual areas in Macaque and human cerebral cortex. In: Chalupa, L., Werner, J.S. (Eds.), *The Visual Neurosciences*. MIT Press, pp. 507–521.
- Van Essen, D.C., Drury, H.A., 1997. Structural and functional analyses of human cerebral cortex using a surface-based atlas. *J. Neurosci.* 17, 7079–7102.
- Van Essen, D.C., Maunsell, J.H.R., 1980. Two-dimensional maps of the cerebral cortex. *J. Comp. Neurol.* 191, 255–281.
- Van Essen, D.C., Dickson, J., Harwell, J., Hanlon, D., Anderson, C.H., Drury, H.A., 2001. An integrated software system for surface-based analyses of cerebral cortex. *J. Am. Med. Inform. Assoc.* 8, 443–459 (Special issue on the Human Brain Project).
- Van Essen, D.C., Snyder, A.Z., Raichle, M.E., Rose, F.E., Bellugi, U., 2004. Differences in cortical shape in Williams syndrome subjects compared to normal humans revealed by surface-based analysis. *Soc. Neurosci. Abstr.* 30, 239.12.
- Van Essen, D.C., Harwell, J., Hanlon, D., Dickson, J., 2005. Surface-based atlases and a database of cortical structure and function. In: Koslow, S.H., Subramaniam, S. (Eds.), *Databasing the Brain, from Data to Knowledge (Neuroinformatics)*. John Wiley and Sons, NJ, pp. 369–388.
- Watkins, K.E., Paus, T., Lerch, J.P., Zijdenbos, A., Collins, D.L., Neelin, P., Taylor, J., Worsley, K.J., Evans, A.C., 2001. Structural asymmetries in the human brain, a voxel-based statistical analysis of 142 MRI scans. *Cereb. Cortex* 11, 868–877.
- Zilles, K., Schleicher, A., Palomero-Gallagher, N., Amunts, K., 2002. Quantitative analysis of cyto- and receptor architecture of the human brain. In: Mazziotta, J.C., Toga, A. (Eds.), *Brain Mapping: The Methods*, pp. 573–602.